Formalisation Report

A proof of the sharpness conjectures
for all saturated fusion systems

Theorem 1.1 and Corollary 1.2 of the accompanying manuscript are formalised in Lean 4 with the Mathlib library and in Rocq with the Mathematical Components library. The definitions, lemmas and propositions of the manuscript also have formal counterparts in Lean and in Rocq, and the counterexample over the rational numbers in item (i) of Section 3.2.1 of this report, concerning the minimal subgroups of simple Mackey functors, is proved in both systems. No result from the literature is assumed in these formal proofs.

Contents

1 The sharpness conjectures and the formal proofs **1**

1.1 The formal definitions 1

1.2 What is proved 2

1.3 References and declaration names 4

1.4 Compilation and checking 4

2 Fusion systems, higher limits and Mackey functors **6**

2.1 Organisation and methods of the proof 6

2.2 The centraliser decomposition 7

2.3 Fusion systems, orbit categories and central subgroups 7

2.4 Higher limits and change of categories 8

2.5 Mackey functors and finite generation 9

2.6 Direct limits of Mackey functors 10

2.7 Simple Mackey functors and minimal subgroups 10

2.8 Restriction to minimal subgroups that do not contain their centralisers 11

2.9 Reduction to simple Mackey functors whose minimal subgroups are not $\mathcal{F}$-centric 12

3 Quotients by central subgroups **13**

3.1 The centric orbit category of the quotient 13

3.2 Mackey functors for the quotient 14

4 Centralisers of elementary abelian subgroups **18**

4.1 Higher limits over the centric orbit categories of the centralisers 18

4.2 Vanishing of the inverse limit 19

4.3 Mackey functors for the centralisers 19

5 The proof of Theorem 1.1 and its consequences **21**

5.1 Induction on the order of the p-group 21

5.2 Coefficients in $\mathbb{Z}_{(p)}$ and group cohomology 21

5.3 Subgroup and normaliser decompositions 22

5.4 The stable elements theorem 23

A Manuscript statements and formal declarations **25**

B Sources used in the manuscript **28**

C Further formal results **32**
C.1 The sharpness of the centraliser decomposition 32
C.2 Other formal proofs of manuscript statements 35
C.3 Long exact sequences of higher limits and composition factors 37

D Logical foundations and sources **38**
D.1 Sources and formal proofs 38
D.2 Formal dependencies 43
D.3 Constructions in the libraries and in the formalisations 44
D.4 The logics of Lean and Rocq 45
D.5 Descriptions of the proof systems and the classical library 46

E Compilation, proof checking and size of the project sources **47**
E.1 Compilation of the project sources 47
E.2 What the checks establish 48
E.3 Time and memory of the checker runs 51
E.4 Size of the project sources 52

Bibliography **53**

Further references **56**

Chapter 1

The sharpness conjectures and the formal proofs

Let p be a prime. Díaz and Park [13, Conjecture] conjectured that, for every saturated fusion system $\mathcal{F}$ over a finite p-group S and every Mackey functor M for $\mathcal{F}$ over $\mathbb{F}_p$ whose values are finite dimensional, the higher limits of the contravariant part M^* of M over the centric orbit category $\mathcal{O}(\mathcal{F}^c)$ vanish in all positive degrees. Theorem 1.1 of the accompanying manuscript is a strengthening of this conjecture to Mackey functors over $\mathbb{Z}_{(p)}$ whose values need not be finitely generated. By Corollary 1.2, for every saturated fusion system $\mathcal{F}$ over a finite p-group and every $n \geqslant 0$, the higher limits of the cohomology functor $H^n(-;\mathbb{F}_p)$ over $\mathcal{O}(\mathcal{F}^c)$ vanish in all positive degrees.

References to manuscript results use the numbering of the accompanying manuscript. The formal counterparts of its numbered statements are given in Appendix A. The formalisations are in Lean with Mathlib and in Rocq with Mathematical Components and MathComp Analysis.

1.1 The formal definitions

Let p be a prime and let $\mathcal{F}$ be a fusion system over a finite p-group S. Its orbit category $\mathcal{O}(\mathcal{F})$ has the subgroups of S as objects, and its morphism sets are $\mathrm{Inn}(R)\backslash \mathrm{Hom}_{\mathcal{F}}(P,R)$ for $P, R \leqslant S$. The centric orbit category $\mathcal{O}(\mathcal{F}^c)$ is the full subcategory of $\mathcal{O}(\mathcal{F})$ whose objects are the $\mathcal{F}$-centric subgroups. Saturation is expressed by the Sylow and extension properties in both formalisations. Aschbacher, Kessar and Oliver [2, Definition I.2.2] call $\mathcal{F}$ saturated if every subgroup of S is $\mathcal{F}$-conjugate to a subgroup that is fully automised and receptive. The relation of the two definitions is explained in Chapter 2.

Central subgroups are defined in Definition 2.2 by an extension condition on the morphisms of $\mathcal{F}$. A Mackey functor consists of a covariant and a contravariant functor on the orbit category, with common values and compatible restriction, transfer and conjugation maps. In both formalisations, conjugation by $s \in S$ sends $x \in S$ to $s^{-1}xs$. The Mackey formula of Definition 2.4 is stated for inclusions. The formal structures also have a form of this formula for all morphisms of $\mathcal{F}$. The two forms of the formula are proved equivalent, and so are the formal definition of central subgroups and that of Definition 2.2. Both equivalences are described in Chapter 2.

Higher limits are the right derived functors of the inverse limit on the category of contravariant functors to modules, and in degree 0 they are the inverse limit. In Lean they are expressed as Ext from the constant functor. In Rocq they are the cohomology of inverse limits of a chosen

injective resolution. The manuscript also describes higher limits as the cohomology groups of the cochain complex induced by the bar resolution. Both the Lean and the Rocq higher limits are proved isomorphic to these groups. The coefficient ring is $\mathbb{Z}_{(p)}$ in Theorem 1.1 and $\mathbb{F}_p$ in Corollary 1.2, and the counterexample in item (i) of Section 3.2.1 has coefficients in $\mathbb{Q}$.

Formalisation.

The Lean constructions use Mathlib's finite groups, quotient groups, module categories, submodules, homological complexes and Ext. The fusion and orbit categories, Mackey functors and the results on higher limits used in the proofs are developed from these notions. The Rocq formalisation uses Mathematical Components' finite groups, quotient groups and modules. The categories of functors and the injective resolutions in them are developed in this formalisation, and the natural isomorphism between the cohomology of the cochain complex induced by the bar resolution and the higher limits is constructed there. The topological consequences of Corollary 1.2 proved in the formalisations are the sharpness of the subgroup and normaliser decompositions and the stable elements theorem, described in Chapter 5. The notions of topology used in their Rocq proofs come from MathComp Analysis, and so does the structure of a real closed field with the Archimedean property and the least upper bound property.

1.2 What is proved

Theorem 1.1 is proved for every saturated fusion system $\mathcal{F}$ over a finite p-group S and every Mackey functor M for $\mathcal{F}$ over $\mathbb{Z}_{(p)}$, with no finiteness condition on its values. Its conclusion is

$$\varprojlim_{\mathcal{O}(\mathcal{F}^c)}{}^{i}\ M^* = 0 \qquad (i > 0).$$

By Corollary 1.2, the same vanishing holds for the cohomology functor $H^n(-;\mathbb{F}_p)$ for every $n \geqslant 0$. Neither result has the hypothesis $S \neq 1$. The saturation of the centralisers and of the central quotients and the vanishing results used in the induction are proved. The Mackey functors for the central quotients in the proof of Proposition 3.2 and the Mackey subfunctors of Lemma 2.6(i) are constructed. No result from the literature is a hypothesis of the formal counterparts of Theorem 1.1 and Corollary 1.2. Their hypotheses and the deduction of Theorem 1.1 over $\mathbb{Z}_{(p)}$ from Proposition 5.1 are given in Chapter 5.

In the supplement on the centraliser decomposition, a centric linking system $\mathcal{L}$ associated to $\mathcal{F}$ is assumed to be given. The declarations

```
Sharpness.FusionSystem.CentricLinkingSystem.
BLO_centralizer_decomposition_sharp
```

and

```
DiazParkBLO.BLOEdgeFinal.DPBLOEdgeFinal.
centralizer_cohomological_sharpness
```

have the hypothesis $S \neq 1$. They state that the higher limits of the centraliser decomposition vanish in positive degrees and that its restriction map from the mod p cohomology of the geometric realisation of $\mathcal{L}$ to the inverse limit is an isomorphism in every degree. The centric linking systems of the centralisers are constructed from $\mathcal{L}$. The existence of a centric linking system for every saturated fusion system is neither a hypothesis nor a conclusion of these declarations. The Rocq statements that involve geometric realisation also take `Real : realType`, a real closed field with the Archimedean property and the least upper bound property. The Rocq formalisation defines

```
DiazParkV16b.TopologicalSimplexReals.DPStandardSimplex.real_line
```

using the real numbers of the Rocq standard library and the module `Rstruct` of MathComp Analysis. These Rocq statements remain quantified over `Real`, and in the Rocq files of the supplement `realType` occurs only in the section variables `Real : realType`. By contrast, the Rocq counterparts of Theorem 1.1 and Corollary 1.2,

```
DiazParkV29.MainManuscript.DPV29DiazParkManuscript.prime_local_main
```

and

```
DiazParkV29.MainCohomology.DPV29GroupCohomologySharpness.
prime_field_higher_limits_zero,
```

are not quantified over `Real`.

The Lean definition of minimal subgroups, `Sharpness.FusionSystem.IsMinimalSupport`, is stated over a field, whereas Definition 2.7 and the Rocq definition

```
DiazPark.MackeyFusionMinimal.DPMackeyMinimal.subgroup_minimal
```

allow a commutative ring. The uses over $\mathbb{F}_p$ are unaffected. Higher limits are proved to commute with colimits over small filtered categories in Lean and over directed posets in Rocq. Several formal statements allow more general coefficients where the manuscript uses $\mathbb{F}_p$. The Rocq counterparts of Lemmas 2.8(i) and 2.8(ii) hold over every commutative ring, those of Lemmas 2.8(iii) and 4.2 over every commutative ring in which $p = 0$, and those of Lemma 2.9 and Propositions 3.2, 4.3 and 5.1 over every field of characteristic p. For Corollary 1.2, the Rocq counterpart concerns cohomology with coefficients in every $\mathbb{F}_p$-module with trivial group action. In Lean, the counterparts of Lemmas 2.8(i) and 2.8(ii) are proved over every field, and those of Lemmas 2.8(iii) and 4.2 over every field of characteristic p.

The formal proofs of Lemma 2.6(ii) use the fact that higher limits commute with filtered colimits. Those of Lemmas 2.8(ii) and 2.8(iii) use the decomposition of a given simple Mackey functor M into images of relative traces. For Lemma 2.8(iii), they also use the vanishing in characteristic p of the relative trace from L to $N_T(L)$ on $M(L)$, for subgroups $L \leqslant T$ of S with $C_T(L) \not\leqslant L$. The formal proofs of Lemma 2.9 use a composition series for Mackey functors whose values are finite dimensional, and they use Lemma 2.6(ii) for Mackey functors whose values need not be finite dimensional. For all four statements, the relation of the formal proofs to the proofs in the manuscript and the differences of generality are explained in Chapter 2. The constructions and lemmas that some of these formal proofs share with other formal proofs of the same statements are given in Appendix C.

The historical account in Section 1 of the manuscript is not formalised, and neither are the homology decompositions of classifying spaces of finite groups recalled there. The homotopy equivalences of the subgroup, normaliser and centraliser decompositions are not among the conclusions of the formal statements on these decompositions, described in Chapter 5 and Section C.1. In particular, the homotopy equivalence of Libman [34] between $|\mathcal{L}|$ and the homotopy colimit of a functor on the poset of $\mathcal{F}$-conjugacy classes of nonempty chains of $\mathcal{F}$-centric subgroups, ordered by reverse inclusion of subchains, is not one of these conclusions. The formal results on the normaliser decomposition concern the vanishing in positive degrees of its higher limits and the invertibility of the restriction map from the cohomology of the geometric realisation $|\mathcal{L}|$ to the inverse limit. In Rocq they also include the collapse of the constructed spectral sequence of a filtered cochain complex. Beyond the kernels, images, cokernels and exact sequences that the formal proofs use, the assertion of Section 2.3 of the manuscript that the Mackey functors form an abelian category is not among the formal conclusions described in this report. Nor are all statements of each cited paper, the cited

theorems in their full generality or a classification of simple Mackey functors for arbitrary coefficient modules. The cited theorems that the formal proofs use are proved in the generality in which the manuscript uses them.

The failure at $S = 1$ of the homotopy equivalence in item (ii) of Section 3.2.1, and the failure of the convergence asserted for the spectral sequence in item (iii), are not established by the inspected formal results. The inspected formal results are the declarations named in this report, whose statements were read in the Lean and Rocq source files. No further theorem on the case $S = 1$ was found in a bounded search of these files. The stable elements theorem and the formal counterparts of Proposition 4.3 do include $S = 1$, as explained in Chapters 3 and 5.

The dependency checks distinguish the proof of Theorem 1.1 from the proofs of the topological consequences of Corollary 1.2. Each check follows the dependencies of selected declarations, called its roots. The roots of the Lean dependency check are 525 declarations of the 28 Lean modules compiled against the other 906, among them the Lean counterparts of Theorem 1.1 and Corollary 1.2 and the theorems on the subgroup and normaliser decompositions. The Rocq dependency check has 16 roots, among them the Rocq counterpart of Theorem 1.1 and the Rocq theorems on the two decompositions. By the methods described in the last section of this chapter, the checks establish use or non-use of named declarations for these roots. In Rocq, the proof that higher limits commute with colimits over directed posets uses both a direct calculation with the cochain complexes induced by the bar resolution and dimension shifting through injective resolutions. The proofs of the topological consequences depend on further results of topology, among them excision for singular homology and the isomorphism between the simplicial homology of a simplicial set and the singular homology of its geometric realisation. In Rocq they also depend on constructed spectral sequences of filtered cochain complexes, their collapse and the formulas for their edge maps.

1.3 References and declaration names

The sources cited in Sections 1–5 of the manuscript are listed, with their locations there, in Appendix B. The bibliography contains 53 entries, in the same order and with the same numbers as entries 1–53 of the manuscript bibliography. The list of further references after the bibliography contains one work outside the manuscript bibliography, labelled [R1]. Numbers and version tags inside formal declaration names and source comments are not references to manuscript numbering. Declaration names keep their own spelling. Table D.1 distinguishes a cited statement from the form proved and used in the formal argument.

1.4 Compilation and checking

The account in this section uses the saved checks of the formalisations and of the separate supplement. The checks with Nanoda, the check with comparator and the size of the project sources are given in Appendix E, and so are the CPU time and memory of the checker runs and of a separate compilation of these sources.

The checks used Lean 4.31.0 and Rocq 9.1.0 with Mathematical Components 2.5.0. The formalisations contain 934 Lean source modules and 1,215 Rocq source modules. The supplement adds 23 and 33, respectively. The compiled files of 906 Lean modules of the formalisations were produced by Lake and by direct calls of the Lean compiler, and the other 28 Lean modules were compiled against these files. In Rocq, 1,173 project modules were rebuilt before the other 42 were compiled. In both systems, the compiled libraries were reused. The additions of the supplement were built against the compiled files of the formalisations.

The Lean kernel checks replayed the selected logical dependency closures, including library declarations, into an initially empty environment. Through the same replay operation, a theorem with a correct proof term was accepted, and the same statement with a term of the wrong type was rejected. These two runs were reused for the supplement. The Rocq checks used `coqchk` on the compiled Rocq files of the project modules named in each invocation and of their recursive dependencies, including those of the libraries. For the Rocq check of the supplement, its compiled files and type queries were reused, and this kernel check was run.

The assumption queries of the formalisations concern 557 Lean and 795 Rocq declarations, and in both systems the queried declarations include every declaration of Table A.1. The supplement has separate assumption queries for 200 Lean and 301 Rocq declarations. For the formalisations, the union of the axioms on which the queried Lean declarations depend is `Classical.choice`, `Quot.sound` and `propext`. The corresponding union for the queried Rocq declarations is `boolp.constructive_indefinite_description`, `boolp.functional_extensionality_dep` and `boolp.propositional_extensionality`. These Rocq axioms are declared in the classical library of MathComp Analysis. In each system, the union for the supplement is the same as for the formalisations. These unions concern the queried declarations, not every declaration in the imported libraries. The module that defines `real_line`, described in Appendix D, is not among the project modules imported, directly or indirectly, by the modules that contain the Rocq counterparts of Theorem 1.1 and Corollary 1.2. These axioms are distinct from the mathematical hypotheses in theorem statements. The sources that the formalisations and the supplement follow are given in Appendix D, where the formal dependencies are traced down to the foundations of formal logic.

The Lean dependency check follows types, proof and definition bodies, and the links between inductive types, their constructors and the computation rules of their recursors in the saved Lean dependency graph, starting from its selected roots. No unresolved reference remains in that graph. The saved Rocq dependency sets are the sets of declarations that the Rocq dependency queries printed for their roots by following proof bodies. A declaration that lies in one of these sets is used, but absence from them does not establish non-use through the types of declarations or through inductive types, their constructors and their recursors. Therefore three outcomes are distinguished for each declaration and each system, namely that it is used, that it is not used and that its use is not established. Table A.1 states only outcomes of the first two kinds, and each of its claims of non-use rests on the saved Lean dependency graph.

The checks rest on trust in the kernels of the two systems, the axioms, the compiler and computing environment, and the integrity of the files. They do not establish that the formal definitions express the intended mathematics. That relation depends on the separate comparisons of statements, definitions and maps described in Chapters 2–5 and Appendix A.

Chapter 2

Fusion systems, higher limits and Mackey functors

Let p be a prime and let $\mathcal{F}$ be a fusion system over a finite p-group S. For fusion systems, the terminology is that of Aschbacher, Kessar and Oliver [2, Part I]. The conventions on formal statements and their hypotheses are those of Chapter 1. The orbit categories and the Mackey functors used in the induction are introduced in Section 2 of the manuscript. By Lemma 2.9, if $\mathcal{F}$ is saturated and $\varprojlim^i_{\mathcal{O}(\mathcal{F}^c)} N^* = 0$ for every $i > 0$ and every simple Mackey functor N for $\mathcal{F}$ over $\mathbb{F}_p$ whose minimal subgroups, in the sense of Definition 2.7, are not $\mathcal{F}$-centric, then $\varprojlim^i_{\mathcal{O}(\mathcal{F}^c)} M^* = 0$ for every $i > 0$ and every Mackey functor M for $\mathcal{F}$ over $\mathbb{F}_p$.

2.1 Organisation and methods of the proof

The proof of Theorem 1.1 begins over $\mathbb{F}_p$, by induction on $|S|$ for all saturated fusion systems. For each nontrivial elementary abelian subgroup E of S that is fully centralised in $\mathcal{F}$, the induction hypothesis applies to $C_{\mathcal{F}}(E)/E$ because $|C_S(E)/E| < |S|$. By Proposition 3.2, the theorem over $\mathbb{F}_p$ then holds for $C_{\mathcal{F}}(E)$, and by Proposition 4.3 it holds for $\mathcal{F}$. The induction concerns all Mackey functors, not only the cohomology functors, since for a central subgroup Z of $\mathcal{F}$ the functor $H^n(-;\mathbb{F}_p)$ need not factor through the functor $P \mapsto P/Z$ from $\mathcal{O}(\mathcal{F}^c)$ to the orbit category of the quotient $\mathcal{F}/Z$. The proof of Proposition 4.3 also needs the vanishing of the inverse limit over $\mathcal{O}(C_{\mathcal{F}}(E)^c)$ for each such E. This vanishing fails for $H^0(-;\mathbb{F}_p)$, but by Lemma 4.2 it holds for simple Mackey functors whose minimal subgroups are not $\mathcal{F}$-centric. The theorem over $\mathbb{Z}_{(p)}$ then follows by Nakayama's lemma and the finite generation of higher limits, first for Mackey functors whose values are finitely generated and then in general by Lemma 2.6(ii).

Broto, Levi and Oliver [6, proof of Theorem 5.8] proved the stable elements theorem by induction through central quotients and centralisers. The argument for the sharpness of the centraliser decomposition in that induction uses a cohomology decomposition theorem of Dwyer and Wilkerson [16, Theorem 1.2] and Lannes's theory of the T-functor [32]. In the proof of Theorem 1.1, the vanishing in Lemma 4.2 takes the place of this sharpness. According to Section 2.1 of the manuscript, that proof uses no case analysis of exotic fusion systems, no classifying space for $\mathcal{F}$ and no infinite group that realises $\mathcal{F}$, and it does not rely on the classification of finite simple groups. The formal proofs of the sharpness of the centraliser decomposition in Section C.1 use geometric realisations and singular cohomology.

Formalisation.

What the dependency checks establish for the formal counterparts of Theorem 1.1 and Corollary 1.2 is described in Chapter 1.

2.2 The centraliser decomposition

For a p-local finite group $(S, \mathcal{F}, \mathcal{L})$ with $S \neq 1$, the centraliser decomposition of Broto, Levi and Oliver [6, Theorem 2.6] is a homotopy equivalence between $|\mathcal{L}|$ and the homotopy colimit of a contravariant functor. The functor is defined on the full subcategory of $\mathcal{F}$ whose objects are the nontrivial elementary abelian subgroups of S that are fully centralised in $\mathcal{F}$. It sends each such subgroup E to a space with the homotopy type of the geometric realisation of the centric linking system of $C_{\mathcal{F}}(E)$ constructed from $\mathcal{L}$ by Broto, Levi and Oliver [6, Definition 2.4 and Proposition 2.5(c)].

Broto, Levi and Oliver [6, proof of Theorem 5.8] proved that this decomposition is sharp. In Section 2.1 of the manuscript, this conclusion is also deduced from Corollary 1.2 and an observation of Yalçın [53, §9, p. 55]. For each nontrivial elementary abelian subgroup E of S fully centralised in $\mathcal{F}$, let $\overline{C}_{\mathcal{L}}(E)$ be the category defined in the statement of the centraliser decomposition and recalled in Section C.1. For every $n \geqslant 0$, it is proved in the supplement on the centraliser decomposition that the higher limits of the functor $E \mapsto H^n(|\overline{C}_{\mathcal{L}}(E)|; \mathbb{F}_p)$ vanish in positive degrees over the opposite of the full subcategory of $\mathcal{F}$ whose objects are these subgroups, and that the canonical map from $H^n(|\mathcal{L}|; \mathbb{F}_p)$ to the inverse limit of this functor, induced by the forgetful functors, is an isomorphism. It is also proved there that the augmentation of the centraliser decomposition, defined in Section C.1, is a quasi-isomorphism. The homotopy equivalence between $|\mathcal{L}|$ and the homotopy colimit is not among these conclusions.

Formalisation.

The declarations `Sharpness.FusionSystem.CentricLinkingSystem.BLO_centralizer_decomposition_sharp` in Lean and `DiazParkBLO.BLOEdgeFinal.DPBLOEdgeFinal.centralizer_cohomological_sharpness` in Rocq each state the vanishing and the isomorphism. Both declarations assume a nontrivial finite p-group, a saturated fusion system and a given centric linking system. The Rocq statement is also quantified over `Real : realType` for geometric realisation. The augmentation of the centraliser decomposition is proved to be a quasi-isomorphism in `Sharpness.FusionSystem.CentricLinkingSystem.BLOEdgeCochainAugmentation_quasiIso` in Lean and `DiazParkBLO.BLOEdgeFinal.DPBLOEdgeFinal.augmentation_cochain_quasi_isomorphism` in Rocq. Neither the existence of a centric linking system for every saturated fusion system nor the homotopy equivalence between $|\mathcal{L}|$ and the homotopy colimit is a hypothesis of these four declarations.

2.3 Fusion systems, orbit categories and central subgroups

In Definition 2.1, a subgroup is $\mathcal{F}$-centric when every $\mathcal{F}$-conjugate contains its centraliser in S. For subgroups $P, R \leqslant S$, the orbit category $\mathcal{O}(\mathcal{F})$ of $\mathcal{F}$ has morphism sets

$$\mathrm{Mor}_{\mathcal{O}(\mathcal{F})}(P, R) = \mathrm{Inn}(R) \backslash \mathrm{Hom}_{\mathcal{F}}(P, R).$$

For a set $\mathcal{V}$ of subgroups closed under $\mathcal{F}$-conjugacy, the full subcategory of $\mathcal{O}(\mathcal{F})$ with object set $\mathcal{V}$ is denoted by $\mathcal{O}^{\mathcal{V}}(\mathcal{F})$. In particular, the centric orbit category is $\mathcal{O}(\mathcal{F}^c) = \mathcal{O}^{\mathcal{F}^c}(\mathcal{F})$. Here $\mathcal{F}^c$ is the set of $\mathcal{F}$-centric subgroups of S, and $\mathcal{F}^{cr}$ consists of those $P \in \mathcal{F}^c$ with $O_p(\mathrm{Out}_{\mathcal{F}}(P)) = 1$. Maps are written on the left and composed from right to left, and for $x \in S$ the map c_x sends $y \in S$ to xyx^{-1}.

In Definition 2.2, a normal subgroup Z of S is central in $\mathcal{F}$ when every morphism $P \to R$ of $\mathcal{F}$ extends to a morphism $PZ \to RZ$ of $\mathcal{F}$ that is the identity on Z. For a subgroup E of S, the centraliser $C_{\mathcal{F}}(E)$ is the fusion system over $C_S(E)$ whose morphisms $P \to R$, for $P, R \leqslant C_S(E)$, are the morphisms in $\mathrm{Hom}_{\mathcal{F}}(P, R)$ that extend to a morphism $PE \to RE$ of $\mathcal{F}$ that is the identity on E. Let $E \leqslant S$ be abelian. Then E is central in $C_{\mathcal{F}}(E)$. Suppose that $\mathcal{F}$ is saturated and E is fully centralised in $\mathcal{F}$. Then $C_{\mathcal{F}}(E)$ is saturated by [6, Proposition 2.5(b)], and Broto, Levi and Oliver [6, Proposition 2.5(a)] showed that a subgroup $P \leqslant C_S(E)$ is $C_{\mathcal{F}}(E)$-centric if and only if $E \leqslant P$ and P is $\mathcal{F}$-centric.

Formalisation.

In Lean, `FusionSystem` and `Saturated` in `Sharpness.FusionSystem` and `Sharpness.Saturation` express the definition of a fusion system and the Sylow and extension properties. In Rocq the corresponding definitions are `DPFusionSystem.system` and `DPFusionSaturation.saturated` in `FusionSystems` and `FusionSaturation`. Aschbacher, Kessar and Oliver [2, Definition I.2.2] define saturation by the condition that every subgroup of S is $\mathcal{F}$-conjugate to a fully automised and receptive subgroup. The Sylow and extension properties used in these formal definitions are equivalent to this definition by [2, Proposition I.2.5], and this equivalence is not formalised. The categories `FusionSystem.Orbit` and `DPFusionOrbit.orbit_category` have as morphisms the morphisms of $\mathcal{F}$ modulo the inner automorphisms of the target. In both systems, conjugation by x sends y to $x^{-1}yx$, so the map c_x of the manuscript is formal conjugation by x^{-1}.

The predicates `CentralSubgroup` and `DPFusionQuotient.central` are proved equivalent to the condition of Definition 2.2 for Z to be central in $\mathcal{F}$ by `centralSubgroup_iff_manuscriptCentralSubgroup` in `Sharpness.V28.CentralSubgroupDefinitions` and `DPV28ManuscriptCentrality.manuscript_central_iff` in `ManuscriptCentrality`. In both systems, $C_{\mathcal{F}}(E)$ is defined for every subgroup E of S. For a finite p-group S, a saturated $\mathcal{F}$ and an abelian subgroup E fully centralised in $\mathcal{F}$, it is proved in `centralizerOfAbelian_saturated_and_central` in `Sharpness.Unconditional.AbelianCentralizerSaturation` and `DPFusionCentralizerSaturation.centralizer_saturated_central` in `FusionCentralizerSaturation` that $C_{\mathcal{F}}(E)$ is saturated and that E is central in $C_{\mathcal{F}}(E)$. Neither result has the hypothesis that E is nontrivial or elementary abelian. Central quotients are discussed in Chapter 3.

2.4 Higher limits and change of categories

Let A be a commutative ring, let $\mathcal{C}$ be a small category and let F be a contravariant functor from $\mathcal{C}$ to A-modules. Its higher limits $\varprojlim^n{}_{\mathcal{C}} F$, for $n \geqslant 0$, are the values at F of the right derived functors of the inverse limit. In degree n, the cochain complex induced by the bar resolution [2, §III.5.1] has one factor $F(c_0)$ for each composable sequence $c_0 \to \cdots \to c_n$ of n morphisms in $\mathcal{C}$. By the dual of [17, Appendix II, Proposition 3.3], its cohomology computes the higher limits. In degree 0 this gives the inverse limit. Let $\Phi\colon \mathcal{D} \to \mathcal{C}$ be a functor of small categories with a left adjoint. By Lemma 2.3, attributed to Jackowski and McClure [30, Lemma 3.1], the higher limits of F over $\mathcal{C}$ and of $F \circ \Phi^{\mathrm{op}}$ over $\mathcal{D}$ are isomorphic in every degree, naturally in F. Jackowski, McClure and Oliver [31, Proposition 5.1(ii)] stated the result for functors to abelian groups. By the naturality of the isomorphism and the invariance of higher limits under a change of rings, it also holds for functors to A-modules.

Formalisation.

In Lean, for small categories, a commutative ring and every $n \geqslant 0$, the natural isomorphism of modules is defined in `higherLimitRightAdjointNatIso` in `Sharpness.Unconditional.HigherLimitAdjunctionNatural` from an adjunction whose right adjoint plays the role of Φ. Its construction uses projective resolutions of the constant functor and the

induced adjunction between the categories of functors. In Rocq, the corresponding natural isomorphism `derived_left_adjoint_nat_iso` in `CategoryDerivedAdjunction` is defined with injective resolutions of covariant functors. Lemma 2.3, which concerns contravariant functors, follows by applying this result to opposite categories, with Φ^{op} as the left adjoint. Both isomorphisms are natural in the coefficients, and the categories need not be finite.

2.5 Mackey functors and finite generation

In Definition 2.4, taken from Díaz and Park [13, Definition 2.1], a Mackey functor $M = (M^*, M_*)$ consists of a contravariant functor M^* and a covariant functor M_* on $\mathcal{O}(\mathcal{F})$ with common values, such that $M^*(\varphi) = M_*(\varphi^{-1})$ for every isomorphism φ of $\mathcal{F}$ and the Mackey formula holds for inclusions. In Definition 2.5, Mackey subfunctors are stable under restriction, transfer and conjugation, and morphisms commute with these maps. Kernels, images and cokernels are formed at each subgroup. Thus a sequence of Mackey functors is exact if and only if its sequence of values at every subgroup is exact.

Let M be a Mackey functor for $\mathcal{F}$ over a commutative ring. For subgroups $P, R \leqslant S$ and $x \in M(P)$, the value at R of the Mackey subfunctor N_x of Lemma 2.6(i) is the submodule of $M(R)$ generated by the elements

$$\mathrm{t}_U^R \,\mathrm{iso}(\alpha)\mathrm{r}_V^P(x), \qquad V \leqslant P, \quad U \leqslant R, \quad \alpha \in \mathrm{Iso}_{\mathcal{F}}(V, U).$$

Here r, t and iso are the restriction, transfer and conjugation maps of M. There are finitely many such triples. For $R = P$, the triple $(V, U, \alpha) = (P, P, \mathrm{Id}_P)$ gives the generator x. Transfer and conjugation send generators to generators. By the Mackey formula, the restriction of a generator is a finite sum of generators of the same form. This proves that N_x is a Mackey subfunctor whose values are finitely generated, without a noetherian hypothesis.

Formalisation.

In both systems, the Mackey structures impose a form of the Mackey formula for any two morphisms of $\mathcal{F}$ with a common target. The structures are `MackeyModule` and `mackey_module`, with Mackey subfunctors `MackeyModule.Subfunctor` and `mackey_subfunctor`. In Lean, the equivalence of this form with the Mackey formula for inclusions is proved in `hasInclusionMackeyFormula_iff_raw` in `Sharpness.Unconditional.InclusionMackeyBridge`. In Rocq, the two implications are proved in `DPMackeyFusion.full_mackey_of_inclusions` in `MackeyFusionBridge` and `DPFusionMackeyFullBridge.full_inclusion_mackey` in `FusionMackeyFullBridge`. Thus the subfunctor conditions apply to both parts of the functor, including transfer.

In Lean, the Mackey subfunctor `finiteGeneratedSubfunctor` in `Sharpness.NormalForms` is defined with restriction followed by transfer along the finitely many pairs of morphisms of $\mathcal{F}$ with a common domain. In each pair, the first morphism maps into P and the second into R. When the two morphisms of a pair are factored through their images, the composite along the pair takes the form $\mathrm{t}_U^R \,\mathrm{iso}(\alpha)\mathrm{r}_V^P$ of the generators of Lemma 2.6(i). In Rocq, it is proved in `DPMackeyFinite.generated_values_finitely_generated` in `MackeyFusionFiniteGeneration` that the values of the Mackey subfunctor generated by a finite family of elements are finitely generated. It is also proved in `DPMackeyNormal.span_to_inclusion_iso_transfer` in `MackeyFusionNormalForms` that restriction followed by transfer along two morphisms of $\mathcal{O}(\mathcal{F})$ with a common domain, the first into P and the second into R, has the form $\mathrm{t}_U^R \,\mathrm{iso}(\alpha)\mathrm{r}_V^P$. These constructions apply to any finite family of generators, any finite group and any commutative coefficient ring. In both systems, the proofs that these Mackey subfunctors are stable under restriction use the Mackey formula for all morphisms of $\mathcal{F}$, and so do the arguments showing that the values of a simple Mackey functor are generated by transfers from the $\mathcal{F}$-conjugates of a minimal subgroup. In Rocq, these arguments and the proof of finite generation rest on one construction of the Mackey subfunctor generated by a family of elements, and on its proofs that these subfunctors are stable under restriction.

2.6 Direct limits of Mackey functors

In the proof of Lemma 2.6(ii), M is the direct limit of its Mackey subfunctors whose values are finitely generated. These subfunctors form a directed system, since the values of the sum of two of them are again finitely generated. By part (i), every element of every value of M lies in one of them. The centric orbit category is finite. Therefore the proof of Aschbacher, Kessar and Oliver [2, Proposition III.5.5(b)] applies, without the noetherian hypothesis of part (a), to the cochain complex induced by the bar resolution for this category. Higher limits commute with this direct limit, so vanishing for all these subfunctors gives vanishing for M.

Formalisation.

In Lean, it is proved in `filteredColimitHigherLimitIso` in `Sharpness.V29.FilteredLimits` that, for functors on a category with finitely many objects and morphisms, higher limits commute with colimits over any small filtered category. The isomorphism between the higher limits of such a colimit and the colimit of the higher limits is natural and agrees with the maps from each stage. It is applied to the directed system of Mackey subfunctors in `fgCentricHigherLimitIso` and `higherLimit_isZero_of_fg_subfunctors_v29` in `Sharpness.V29.FilteredLimitsMackey`.

In Rocq, the corresponding isomorphism for colimits over directed posets is constructed in `DPFilteredDerived.higher_limit_colimit_iso` in `FilteredLimitsDerived`, and in `DPFilteredLimitsMackey.centric_subfunctor_higher_limit_closure` in `CompositionFilteredMackey` it is applied to the Mackey subfunctors of M whose values are finitely generated, ordered by inclusion. Both proofs use the finite products in each degree of the cochain complex induced by the bar resolution, the exactness of filtered colimits and the natural isomorphism between the cohomology of this complex and the higher limits. The results hold over every commutative ring, for arbitrary module values and in every degree $n \geqslant 0$. No saturation, noetherian or finite dimensionality hypothesis is added. These proofs are used by the formal proofs of Theorem 1.1 in both systems.

2.7 Simple Mackey functors and minimal subgroups

In Definition 2.7, a nonzero Mackey functor is called simple when it has no proper nonzero Mackey subfunctor, and a minimal subgroup for a Mackey functor is a subgroup that is minimal under inclusion among the subgroups at which the value of the functor is nonzero. Let M be a simple Mackey functor for $\mathcal{F}$ over $\mathbb{F}_p$. By Lemma 2.8(i), its minimal subgroups form one $\mathcal{F}$-conjugacy class, and $M(L)$ is a simple $\mathbb{F}_p \operatorname{Out}_{\mathcal{F}}(L)$-module for each minimal subgroup L. Díaz and Park [13, Proposition 3.1] stated this result and noted that Webb's proof [52, Proposition 2.1] for finite groups also applies to fusion systems.

For part (ii), let Q be a minimal subgroup for M and let $U \leqslant S$. By part (i), $M(Q)$ is a simple $\mathbb{F}_p \operatorname{Out}_{\mathcal{F}}(Q)$-module, and by [13, Proposition 3.2], M is isomorphic to the simple Mackey functor $S_{Q,M(Q)}$ that Díaz and Park [13, §3] constructed from Q and $M(Q)$. Under this isomorphism, by [13, formula (4)], $M(U)$ is a direct sum of images of relative traces, one for each U-conjugacy class of $\mathcal{F}$-conjugates L of Q contained in U. By [13, formula (6)], restriction to a representative L of such a class is injective on the summand for L and zero on the other summands. If every one of these restrictions vanishes on an element of $M(U)$, the element is zero. If no $\mathcal{F}$-conjugate of Q lies in U, the direct sum is zero.

Formalisation.

In Lean, `MackeyModule.IsSimple` is defined over an arbitrary ring. Unlike Definition 2.7, which is formulated over a commutative ring, `IsMinimalSupport` in `Sharpness.Unconditional.MinimalSupport` is defined over a field. This predicate expresses minimality under inclusion, and part (i) is proved for any field in `simple_minimalSupport_class` in `Sharpness.Manuscript.SimpleStatements`. In Rocq, `mackey_simple`,

`DPMackeyMinimal.subgroup_minimal`, `DPMackeyMinimalManuscript.minimal_support_iff_conjugate` and `minimal_outer_group_algebra_simple` are formulated over any commutative ring.

For a given simple Mackey functor M and any minimal subgroup Q for M, it is proved in `simpleFunctorAtMinimalIso` in `Sharpness.V29.SimpleFunctor` and `DPSimpleFunctor.simple_functor_identification` in `SimpleFunctor` that M is isomorphic to a Mackey functor constructed from $M(Q)$. Each value of this Mackey functor is a direct sum of images of relative traces. The coefficient ring is a field in Lean and may be any commutative ring in Rocq. In both systems, for each $U \leqslant S$, a representative L is chosen in each U-conjugacy class of $\mathcal{F}$-conjugates of Q contained in U, together with an isomorphism from Q to L in $\mathcal{O}(\mathcal{F})$. It is proved that $M(U)$ is the direct sum of one summand for each representative L, and that $M(U) = 0$ when no $\mathcal{F}$-conjugate of Q lies in U. Under the isomorphism between $M(U)$ and the value at U of the Mackey functor constructed from $M(Q)$, the summand for L corresponds to a submodule of $M(Q)$. The conjugation map from $M(Q)$ to $M(L)$ induced by the isomorphism chosen for L carries this submodule onto the image of the relative trace from L to $N_U(L)$ in $M(L)$. Restriction to L maps the summand for L isomorphically onto this image and is zero on the other summands. For $u \in U$, restriction to uLu^{-1} is restriction to L followed by the conjugation map of M induced by c_u. Hence restriction to uLu^{-1} is also zero on the summands other than that for L. The conclusion of Lemma 2.8(ii), that an element of $M(U)$ is zero when its restrictions to all $\mathcal{F}$-conjugates of Q contained in U vanish, is proved in `simple_functor_restriction_detection_v29` in `Sharpness.V29.SimpleRestrictionFormula` and `DPSimpleFunctor.simple_functor_restriction_detection` in `SimpleFunctor`. These results do not classify functors constructed from arbitrary coefficient modules. The proofs use the facts that the elements whose restrictions to the $\mathcal{F}$-conjugates of Q vanish form a Mackey subfunctor and that the values of M are generated by transfers from these conjugates. They also rest on two identities, which follow from the Mackey formula because the value of M is zero at every subgroup of order less than $|Q|$. For $U \leqslant S$ and $\mathcal{F}$-conjugates L and K of Q contained in U, restriction to L after transfer from L is the relative trace from L to $N_U(L)$ on $M(L)$. Restriction to L after transfer from K is zero if K is not U-conjugate to L. The decomposition into images of relative traces and the conclusion of Lemma 2.8(ii) are used by the formal proofs of Theorem 1.1 in both systems.

2.8 Restriction to minimal subgroups that do not contain their centralisers

Over $\mathbb{F}_p$, Grazian and Marmo [21, Lemma 2.9] and Díaz and Park [13, Corollary 3.4] gave analogues for fusion systems of a result of Webb [52, Proposition 2.8]. In Lemma 2.8(iii) of the manuscript, $\mathcal{F}$ is saturated and M is a simple Mackey functor for $\mathcal{F}$ over $\mathbb{F}_p$. Let $L \leqslant T \leqslant S$, where L is $\mathcal{F}$-conjugate to a minimal subgroup for M and $C_T(L) \not\leqslant L$. Choose L to represent its T-conjugacy class in the decomposition of $M(T)$ in the proof of part (ii). By the lemma of Grazian and Marmo, the summand for L vanishes. Restriction to L is zero on the other summands. Hence $\mathrm{r}_L^T = 0$.

Formalisation.

In Lean, this vanishing is proved in `simple_centralizer_zero_v29` in `Sharpness.V29.SimpleCentralizer` for a finite p-group and a field of characteristic p. In `Sharpness.V29.SimpleCentralizerTrace`, the relative trace from L to $N_T(L)$ on $M(L)$ is proved to be zero by means of the subgroup $LC_T(L)$ of $N_T(L)$. The quotient $LC_T(L)/L$ is a nontrivial p-group, and it acts trivially on $M(L)$ because its image in $\mathrm{Out}_{\mathcal{F}}(L)$ is trivial. Therefore the relative trace from L to $LC_T(L)$ on $M(L)$ is multiplication by the index $|LC_T(L) : L|$, and this map is zero in characteristic p. By the transitivity of relative traces, the relative trace from L to $N_T(L)$ factors through this map, and so it is zero. It follows that the summand for the chosen representative of the T-conjugacy class of L vanishes. As restriction to L is zero on the other summands, $\mathrm{r}_L^T = 0$.

In Rocq, the relative trace from L to $N_T(L)$ on $M(L)$ is also proved to be zero by means of the subgroup $LC_T(L)$, in `SimpleTraceSummands`, `TraceCentralizerIndex` and `TraceCentralizerZero`, and

`DPSimpleManuscript.simple_functor_centralizer_restriction_zero` in `SimpleManuscript` is proved from this vanishing. Its commutative coefficient ring satisfies $p \cdot 1 = 0$, and the subgroup T is a p-group. Neither proof needs saturation. The hypotheses of Lemma 2.8(iii) imply those of both declarations. Both main proofs use the vanishing of the relative trace from L to $N_T(L)$ on $M(L)$ and the restriction formulas of the summands. The proof of this vanishing uses $p \cdot 1 = 0$, independently of the more general ring formulation of simplicity.

2.9 Reduction to simple Mackey functors whose minimal subgroups are not $\mathcal{F}$-centric

Suppose that $\mathcal{F}$ is saturated, and let M be a Mackey functor for $\mathcal{F}$ over $\mathbb{F}_p$. In the proof of Lemma 2.9, by Lemma 2.6(ii) it suffices to consider the case in which the values of M are finite dimensional. Díaz and Park [13, Proposition 4.3] showed that it then suffices to prove that $\varprojlim^i{}_{\mathcal{O}(\mathcal{F}^c)} N^* = 0$ for every $i > 0$ and every simple composition factor N of M with a minimal subgroup that is not $\mathcal{F}$-centric. By Lemma 2.8(i), all minimal subgroups for such a factor are $\mathcal{F}$-conjugate, so none is $\mathcal{F}$-centric and the hypothesis of Lemma 2.9 applies.

Formalisation.

In Lean, it is proved in `finite_dimensional_noncentric_factors_reduction` in `Sharpness.V29.ReductionComposition` that, when the values of M are finite dimensional, M has a composition series with the following property. If $\varprojlim^i{}_{\mathcal{O}(\mathcal{F}^c)} N^* = 0$ for every $i > 0$ and every composition factor N of this series with a minimal subgroup that is not $\mathcal{F}$-centric, then $\varprojlim^i{}_{\mathcal{O}(\mathcal{F}^c)} M^* = 0$ for every $i > 0$. In `V29.noncentric_simple_reduction` in `Sharpness.V29.ReductionArbitrary`, the case of Mackey functors over $\mathbb{F}_p$ whose values need not be finite dimensional then follows by Lemma 2.6(ii). The hypotheses of this declaration are a finite p-group, a saturated fusion system and the vanishing of $\varprojlim^i{}_{\mathcal{O}(\mathcal{F}^c)} N^*$ for every $i > 0$ and every simple Mackey functor N for $\mathcal{F}$ over $\mathbb{F}_p$ whose minimal subgroups are not $\mathcal{F}$-centric.

In Rocq, the corresponding results over any field of characteristic p are proved in `DPCompositionReduction.finite_noncentric_composition_factor_reduction` in `CompositionReduction` and `DPV29MackeyReduction.noncentric_simple_to_arbitrary_higher_limits` in `ReductionManuscript`. For each short exact sequence of Mackey functors in the composition series, the Rocq proof uses the long exact sequence of higher limits, `DPCompositionExact.subquotient_higher_limit_long_exact`. The Lean proof uses `composition_extension_acyclic`, by which the higher limits over $\mathcal{O}(\mathcal{F}^c)$ of the contravariant part of a Mackey functor vanish in positive degrees whenever those of a Mackey subfunctor and of the corresponding quotient vanish. Its proof rests on `higherLimit_map_isIso_of_kernel_acyclic`, the statement that a surjective map of functors whose kernel has zero higher limits in positive degrees induces isomorphisms of higher limits in those degrees. The long exact sequence of higher limits for a short exact sequence of Mackey functors, in `Sharpness.V29.CompositionDerivedLES`, is also proved, but is not used by the Lean formal proof of Theorem 1.1. No declaration of that module is reached from `Sharpness.FusionSystem.V29.primeLocal_sharpness`, the Lean counterpart of Theorem 1.1, in the saved graph of the Lean dependency check. In both systems, the formal proofs of Lemmas 2.6(ii) and 2.9 depend on the natural isomorphism between the cohomology of the cochain complex induced by the bar resolution and the higher limits. Those of Lemma 2.9 also use the vanishing of $\varprojlim^i{}_{\mathcal{O}(\mathcal{F}^c)} N^*$ for every $i > 0$ and every simple Mackey functor N for $\mathcal{F}$ whose minimal subgroups are $\mathcal{F}$-centric. The formal proofs of Theorem 1.1 use the formal proofs of Lemmas 2.6(ii), 2.8(ii), 2.8(iii) and 2.9 described in this chapter.

Chapter 3

Quotients by central subgroups

Let $\mathcal{F}$ be a saturated fusion system over a finite p-group S, and let Z be central in $\mathcal{F}$. Then $Z \leqslant Z(S)$, and every morphism $\varphi\colon P \to R$ of $\mathcal{F}$ between subgroups containing Z is the identity on Z and induces an injective homomorphism $\overline{\varphi}\colon P/Z \to R/Z$. The quotient $\mathcal{F}/Z$ of $\mathcal{F}$ by Z [2, Definition II.5.3] is the fusion system over S/Z whose morphisms are these homomorphisms $\overline{\varphi}$. It is saturated by [2, Lemma II.5.5]. To compare higher limits over the centric orbit categories of $\mathcal{F}$ and $\mathcal{F}/Z$, the manuscript uses the subgroups P containing Z for which P/Z is $\mathcal{F}/Z$-centric. These subgroups need not be all the centric subgroups of $\mathcal{F}$, but they include the centric radical subgroups.

For a simple Mackey functor for $\mathcal{F}$ over $\mathbb{F}_p$, the proof of Proposition 3.2 treats two cases, according as a minimal subgroup contains Z or does not. In the first case the functor gives a Mackey functor for $\mathcal{F}/Z$ with the same values. In the second case it vanishes on $\mathcal{F}$-centric subgroups. The conventions on formal statements and their hypotheses are those of Chapter 1.

3.1 The centric orbit category of the quotient

As in Lemma 3.1, let $\mathcal{X}$ be the set of subgroups $P \leqslant S$ that contain Z and for which P/Z is $\mathcal{F}/Z$-centric. Closure under $\mathcal{F}$-conjugacy and overgroups follows from the corresponding closure properties of centric subgroups in $\mathcal{F}/Z$. By [4, Lemma 6.4(a)], $\mathcal{X} \subseteq \mathcal{F}^c$. The proof of [4, Corollary 6.14, p. 3856] shows that $O_p(\mathrm{Out}_{\mathcal{F}}(P)) \neq 1$ for every $\mathcal{F}$-centric subgroup P outside $\mathcal{X}$, and hence that $\mathcal{F}^{cr} \subseteq \mathcal{X}$.

The quotient maps give an isomorphism from $\mathcal{O}^{\mathcal{X}}(\mathcal{F})$ to $\mathcal{O}((\mathcal{F}/Z)^c)$. On objects it sends P to P/Z, and on morphisms it sends the class represented by φ to that represented by $\overline{\varphi}$. This is the isomorphism in [4, p. 3856, equation (3)]. In parts (i) and (ii), the proof of Lemma 3.1 uses arguments from the proof of the converse in [4, Corollary 6.14]. That proof assumes that $\mathcal{F}/Z$ is the fusion system of a finite group, but these arguments require only that $\mathcal{F}$ is saturated and Z is central in $\mathcal{F}$.

Formalisation.

In Lean, the closure properties and the two inclusions of Lemma 3.1(i) are proved by `Sharpness.FusionSystem.quotientCollection_conjugate`, `Sharpness.FusionSystem.quotientCollection_overgroup`, `Sharpness.FusionSystem.quotientCollection_centric` and `Sharpness.FusionSystem.centric_radical_in_quotientCollection` in `Sharpness.QuotientCollection`. In these declarations, the quotient $\mathcal{F}/Z$ is given by the structure constructed by `Sharpness.FusionSystem.akoQuotientInput`, and its saturation is proved by `Sharpness.FusionSystem.centralQuotient_saturated` in

`Sharpness.Unconditional.QuotientSaturationComplete`. Thus saturation of the quotient is proved from saturation of $\mathcal{F}$. The isomorphism of categories is `Sharpness.FusionSystem.quotientActualCentricIsoCatUnconditional` in `Sharpness.Manuscript.QuotientIsomorphisms`. Its formulas on objects and on morphisms are proved in the same module by `Sharpness.FusionSystem.quotientActualCentricIsoCatUnconditional_obj` and `Sharpness.FusionSystem.quotientActualCentricIsoCatUnconditional_forward`.

In Rocq, the closure properties and the two inclusions of Lemma 3.1(i) are proved by `DPFusionQuotientCollectionClosure.collection_conjugate` and `DPFusionQuotientCollectionClosure.collection_overgroup` in `DiazPark.FusionQuotientCollectionClosure`, `DPFusionQuotientCollection.collection_centric` in `DiazPark.FusionQuotientCollection`, and `DPFusionQuotientRadical.centric_radical_collection` in `DiazPark.FusionQuotientRadical`. Saturation is `DPFusionQuotientSaturation.central_quotient_saturated` in `DiazPark.FusionQuotientSaturation`. The isomorphism and its object and morphism formulas are `DPDiazParkManuscript.quotient_collection_category_isomorphism`, `DPDiazParkManuscript.quotient_collection_object` and `DPDiazParkManuscript.quotient_collection_map` in `DiazParkV16b.FusionSharpnessManuscript`. In both systems, an isomorphism of categories is proved, with mutually inverse functors, rather than only an equivalence.

3.2 Mackey functors for the quotient

By Proposition 3.2, if $\varprojlim^i{}_{\mathcal{O}((\mathcal{F}/Z)^c)} N^* = 0$ for every $i > 0$ and every Mackey functor N for $\mathcal{F}/Z$ over $\mathbb{F}_p$, then $\varprojlim^i{}_{\mathcal{O}(\mathcal{F}^c)} M^* = 0$ for every $i > 0$ and every Mackey functor M for $\mathcal{F}$ over $\mathbb{F}_p$. By Lemma 2.9, it suffices to consider a simple M and a minimal subgroup Q. As Z is central in $\mathcal{F}$, either all the $\mathcal{F}$-conjugates of Q contain Z or none does.

Suppose $Z \not\leqslant Q$. Every $\mathcal{F}$-centric subgroup P contains Z. If $L \leqslant P$ is $\mathcal{F}$-conjugate to Q, then $Z \leqslant C_P(L)$ but $Z \not\leqslant L$. By Lemma 2.8(iii), the restriction from $M(P)$ to $M(L)$ is zero. Thus $M(P) = 0$ by Lemma 2.8(ii). Hence M^* vanishes on $\mathcal{O}(\mathcal{F}^c)$.

Now suppose $Z \leqslant Q$. For $P \geqslant Z$, let Γ_P be the kernel of the map from $\mathrm{Aut}_{\mathcal{F}}(P)$ to $\mathrm{Aut}_{\mathcal{F}/Z}(P/Z)$. As a kernel, Γ_P is normal in $\mathrm{Aut}_{\mathcal{F}}(P)$. Its action on Z and P/Z is trivial, so it is a p-group (see, for example, [20, Corollary 5.3.3]). For each $\mathcal{F}$-conjugate L of Q, the simple $\mathbb{F}_p\,\mathrm{Out}_{\mathcal{F}}(L)$-module $M(L)$ has trivial Γ_L-action. By Lemma 2.8(ii), Γ_P acts trivially on $M(P)$ for every $P \geqslant Z$.

Every morphism of $\mathcal{F}/Z$ is $\overline{\psi}$ for some lift $\psi\colon P \to R$ in $\mathcal{F}$ between subgroups containing Z. The Mackey functor N for $\mathcal{F}/Z$ is defined by $N(P/Z) = M(P)$, $N^*(\overline{\psi}) = M^*(\psi)$ and $N_*(\overline{\psi}) = M_*(\psi)$. If two lifts $\psi, \chi\colon P \to R$ induce the same homomorphism $P/Z \to R/Z$, their images coincide since both contain Z. Thus $\chi = \psi\gamma$ for some $\gamma \in \Gamma_P$. Both $M_*(\gamma)$ and $M^*(\gamma)$ are the identity of $M(P)$, so $M_*(\chi) = M_*(\psi)$ and $M^*(\chi) = M^*(\psi)$. For $y \in R$, the morphism $c_{yZ}|_{R/Z}\,\overline{\psi}$ of $\mathcal{F}/Z$ is induced by $c_y|_R\,\psi$, and $c_y|_R\,\psi$ represents the same morphism of $\mathcal{O}(\mathcal{F})$ as ψ. Finally, taking quotients by Z preserves the indexing by double cosets and the intersections in the Mackey formula. Consequently, for subgroups $U, W \leqslant P \leqslant S$ that contain Z, the Mackey formula for M and U, W agrees term by term with that for N and $U/Z, W/Z$, and N is a Mackey functor for $\mathcal{F}/Z$.

By Lemma 3.1(i), $\mathcal{X}$ contains $\mathcal{F}^{cr}$ and is closed under $\mathcal{F}$-conjugacy and overgroups. For such a set, Broto, Levi and Oliver [6, proof of Corollary 3.6] showed, using only that $\mathcal{F}$ is saturated, that restriction from $\mathcal{O}(\mathcal{F}^c)$ to $\mathcal{O}^{\mathcal{X}}(\mathcal{F})$ preserves higher limits for $\mathbb{Z}_{(p)}$-module

coefficients. Regarding the values of M^* as $\mathbb{Z}_{(p)}$-modules leaves their higher limits unchanged. By the definition of N, the isomorphism of categories of Lemma 3.1(ii) and the hypothesis of Proposition 3.2 applied to N,

$$\varprojlim_{\mathcal{O}(\mathcal{F}^c)}{}^{i}\ M^* \cong \varprojlim_{\mathcal{O}^{\mathcal{X}}(\mathcal{F})}{}^{i}\ M^* \cong \varprojlim_{\mathcal{O}((\mathcal{F}/Z)^c)}{}^{i}\ N^* = 0 \qquad (i > 0).$$

Formalisation.

In Lean, Proposition 3.2 is proved by `Sharpness.FusionSystem.V29.central_quotient_sharpness` in `Sharpness.V29.MainCentralQuotient` for every Mackey functor over $\mathbb{F}_p$, with no finiteness condition on its values. The formal proof has the two cases above. The case $Z \not\leqslant Q$ uses `Sharpness.FusionSystem.V29.simple_centric_zero_of_central_missing`. For $Z \leqslant Q$, the Mackey functor N for $\mathcal{F}/Z$ is defined by `Sharpness.FusionSystem.quotientMackey` in `Sharpness.QuotientMackey`, and its Mackey formula is proved there for all morphisms of $\mathcal{F}/Z$. The contravariant part N^* of N and the functor on $\mathcal{O}(\mathcal{F}/Z)$ defined directly from M^* through lifts are separate formal definitions with the same values and maps. The isomorphism between them whose components are identity maps is `Sharpness.FusionSystem.quotientMackeyCoefficientIso`. The higher limits of a functor on $\mathcal{O}(\mathcal{F}^c)$ are proved to vanish in positive degrees if and only if those of its restriction to $\mathcal{O}^{\mathcal{X}}(\mathcal{F})$ do. The proof of this equivalence uses the vanishing of the higher limits of every functor on $\mathcal{O}(\mathcal{F}^c)$ that is zero outside the $\mathcal{F}$-conjugacy class of an $\mathcal{F}$-centric subgroup P with $O_p(\mathrm{Out}_{\mathcal{F}}(P)) \neq 1$. This vanishing is also used in the proof by Broto, Levi and Oliver that restriction from $\mathcal{O}(\mathcal{F}^c)$ to $\mathcal{O}^{\mathcal{X}}(\mathcal{F})$ preserves higher limits for $\mathbb{Z}_{(p)}$-module coefficients. Neither the equivalence nor the vanishing is assumed from the literature.

In Rocq, Proposition 3.2 is proved by `DPV29CentralQuotientSharpness.central_quotient_sharpness` in `DiazParkV29.MainCentralQuotient` over every field of characteristic p, again with no finiteness condition on the values. The formal proof uses `DPV29CentralQuotientCore.missing_centre_value_zero` and `DPV29CentralQuotientCore.paired_simple_central_quotient_sharpness` in `DiazParkV29.MainCentralQuotientCore`. For $Z \leqslant Q$, the Mackey functor N for $\mathcal{F}/Z$ is defined by `DPFusionMackeyQuotientDescent.simple_quotient` in `DiazPark.FusionMackeyQuotientDescent`. The theorem `DPFusionCentralQuotientSharpness.containing_simple_collection_zero` in `DiazPark.FusionCentralQuotientSharpness` assumes $Z \leqslant Q$ and states that the higher limits of M^* over $\mathcal{O}^{\mathcal{X}}(\mathcal{F})$ vanish in positive degrees. Its proof uses the fact that the restriction of M^* to $\mathcal{O}^{\mathcal{X}}(\mathcal{F})$ is the composite of N^* with the isomorphism of categories of Lemma 3.1(ii). The vanishing over $\mathcal{O}(\mathcal{F}^c)$ then follows from the theorem that the higher limits of a functor on $\mathcal{O}(\mathcal{F}^c)$ vanish in positive degrees if and only if those of its restriction to $\mathcal{O}^{\mathcal{X}}(\mathcal{F})$ do. The hypothesis on $\mathcal{F}/Z$ is the hypothesis of Proposition 3.2 itself, whereas the quotient $\mathcal{F}/Z$ is constructed, and its saturation and these results on restriction to $\mathcal{O}^{\mathcal{X}}(\mathcal{F})$ are proved, in both systems. The construction of N uses the fact that Γ_P acts trivially on $M(P)$ for every $P \geqslant Z$, which the manuscript deduces from Lemma 2.8(ii). Since the modules in which N is defined in the two systems import, directly or indirectly, no module that contains the formal counterparts of Lemma 2.8(ii), the formal constructions of N do not use those counterparts.

3.2.1 Corrections to the literature

Let p be a prime, let S be a finite p-group and let $\mathcal{F}$ be a saturated fusion system over S.

(i) The statement of [13, Corollary 3.4] of Díaz and Park allows an arbitrary commutative coefficient ring. It implies that if a simple Mackey functor is nonzero at $P \leqslant S$, then some minimal subgroup $L \leqslant P$ satisfies $C_P(L) \leqslant L$. For $S \neq 1$, the Mackey functor $P \mapsto H^0(P; \mathbb{Q})$ is a counterexample. Its restriction and isomorphism maps are identities, and transfer from Q to P is multiplication by $[P : Q]$. It is simple, has minimal subgroup 1

and is nonzero at S, although $C_S(1) \not\leqslant 1$. Indeed, a nonzero Mackey subfunctor has value $\mathbb{Q}$ at some subgroup; restriction to 1 and transfer from 1 to any subgroup then force its value to be $\mathbb{Q}$ everywhere. The proof of Lemma 2.8(iii) of the manuscript instead uses Grazian and Marmo [21, Lemma 2.9] over $\mathbb{F}_p$. The applications of the corollary in the proof of [13, Theorem C] of Díaz and Park also have coefficients in $\mathbb{F}_p$. Both Proposition 2.8 of Webb [52] and [13, Proposition 3.6] assume a field of characteristic p. The manuscript uses Lemma 2.8(iii) only in Proposition 3.2 and Lemma 4.2.

(ii) For a p-local finite group $(S, \mathcal{F}, \mathcal{L})$ with $S \neq 1$, the centraliser decomposition expresses $|\mathcal{L}|$ as a homotopy colimit indexed by the opposite of the full subcategory of $\mathcal{F}$ on the nontrivial elementary abelian subgroups of S that are fully centralised in $\mathcal{F}$. In [6, Theorem 2.6] and [2, Theorem III.5.35] this decomposition is stated without the hypothesis $S \neq 1$. When $S = 1$, the indexing category is empty, so the homotopy colimit is empty, whereas $|\mathcal{L}|$ is a point. The manuscript therefore describes this decomposition only for $S \neq 1$. For $S = 1$, Case 2 of the proof of Broto, Levi and Oliver [6, Theorem 5.8] uses both this decomposition and the assertion that S has nontrivial centre. The stable elements theorem itself holds directly in this case, since $|\mathcal{L}|$ is a point and the centric orbit category has only one morphism.

(iii) In Section 4 the manuscript uses the centraliser spectral sequence of Yalçın [53, Theorem 9.4] only when $S \neq 1$. Yalçın states it for every saturated fusion system. When $S = 1$, its indexing category is empty and every E_2-term is zero. For the constant functor with value $\mathbb{F}_p$, however, the limit over the centric orbit category is $\mathbb{F}_p$. Thus the asserted convergence requires $S \neq 1$. Yalçın [53, Theorem 1.8] also uses this spectral sequence without excluding $S = 1$ in the proof of the reduction to fusion systems with nontrivial centre. That reduction remains valid for $S = 1$: the centric orbit category has only one morphism, so its higher limits vanish in positive degrees.

Formalisation.

For item (i), the constructions in both systems start from group cohomology with trivial rational coefficients. In Lean, `Sharpness.FusionSystem.rationalDegreeZeroMackey` in `Sharpness.V29.RationalDegreeZero` is the Mackey functor $P \mapsto H^0(P;\mathbb{Q})$. The isomorphism of each value with $\mathbb{Q}$ is `rationalDegreeZeroCoordinate`, and the formulas for the restriction, transfer and conjugation maps of this Mackey functor are proved by `rationalDegreeZero_pull`, `rationalDegreeZero_transfer` and `rationalDegreeZero_conjugation` in that namespace. Simplicity, the unique minimal subgroup 1, nonzero values and the counterexample are proved in the same module by `rationalDegreeZero_simple`, `rationalDegreeZero_minimal_iff`, `rationalDegreeZero_nonzero` and `rationalDegreeZero_centralizer_counterexample`. The last theorem assumes $S \neq 1$ and states that no minimal subgroup L satisfies $C_S(L) \leqslant L$ although the value at S is nonzero. The example on the group of order 2 is `Sharpness.FusionSystem.rationalExample_counterexample` in `Sharpness.V29.RationalExample`. Its saturation theorem, `Sharpness.FusionSystem.rationalExample_saturated`, uses `Sharpness.finiteGroupFusion_saturated` in `Sharpness.Unconditional.FiniteGroupFusionSaturation`.

In Rocq, `DPV29RationalMackey.rational_mackey` in `DiazParkV29.RationalMackey` is also the Mackey functor $P \mapsto H^0(P;\mathbb{Q})$. The isomorphism of each value with $\mathbb{Q}$ is `DPV29RationalDegreeZero.degree_zero_iso` in `DiazParkV29.RationalDegreeZero`, and the formulas for the maps of this Mackey functor are proved by `DPV29RationalMackey.pull_coordinate`, `DPV29RationalMackey.transfer_coordinate` and `DPV29RationalMackey.conjugation_coordinate`. Simplicity, the unique minimal subgroup 1, nonzero values and the counterexample for $S \neq 1$ are `DPV29RationalMackey.rational_mackey_simple`, `DPV29RationalMackey.rational_minimal_iff`, `DPV29RationalMackey.rational_value_nonzero` and

`DPV29RationalMackey.rational_centralizer_counterexample`. The saturated example on the group of order 2 is `DPV29RationalExample.example_counterexample`, with `DPV29RationalExample.example_saturated`, in `DiazParkV29.RationalExample`, and saturation is proved using `DPFiniteGroupFusionSaturation.saturated` in `DiazParkV16b.FiniteGroupFusionSaturation`. Thus the counterexample is proved in both systems, separately from the results over $\mathbb{F}_p$.

Formalisation.

For item (ii), the failure of the asserted homotopy equivalence at $S = 1$ is not established by the inspected formal results. The main statements of the supplement on the centraliser decomposition, described in Appendix C, have the hypothesis $S \neq 1$ and do not establish that failure. The stable elements theorem does cover $S = 1$ in both systems. Neither `Sharpness.FusionSystem.CentricLinkingSystem.V29.nativeOrdinaryRestrictionIso` in `Sharpness.V29.MainSubgroup` nor `DPV29LinkingSubgroup.native_edge_iso` in `DiazParkV29.MainSubgroup` assumes $S \neq 1$, so each applies in this case as well.

Formalisation.

For item (iii), the failure at $S = 1$ of the convergence asserted for the spectral sequence is likewise not established by the inspected formal results. The formal counterparts of Proposition 4.3 do cover $S = 1$. In Lean, the proof of `Sharpness.FusionSystem.V29.centralizer_sharpness_of_local` in `Sharpness.V29.MainCentralizer` treats the case $S = 1$ separately. In Rocq, `DPV29DiazPark.centralizer_criterion` in `DiazParkV29.MainField` has no nontriviality hypothesis, and its reduction to noncentric minimal subgroups excludes any such subgroup when $S = 1$. These results concern all Mackey functors over $\mathbb{F}_p$ in Lean and every field of characteristic p in Rocq, and hence include the cohomology functors in the reduction cited in item (iii).

Chapter 4

Centralisers of elementary abelian subgroups

4.1 Higher limits over the centric orbit categories of the centralisers

Let $\mathcal{F}$ be a saturated fusion system over a finite p-group $S \neq 1$, and let $\mathcal{F}^e$ be the full subcategory of $\mathcal{F}$ whose objects are the nontrivial elementary abelian subgroups of S that are fully centralised in $\mathcal{F}$. Lemma 4.1 concerns an arbitrary functor $F\colon \mathcal{O}(\mathcal{F}^c)^{\mathrm{op}} \to A\text{-}\mathtt{mod}$, where A is a commutative ring. If its restriction to $\mathcal{O}(C_{\mathcal{F}}(E)^c)$ has zero higher limits in every degree $n \geqslant 0$ for every E in $\mathcal{F}^e$, then the higher limits of F over $\mathcal{O}(\mathcal{F}^c)$ vanish in every degree. The hypothesis includes the inverse limit itself.

The proof uses the category $\mathcal{P}$ of pairs (P, E) with $P \in \mathcal{F}^c$ and $1 \neq E \leqslant Z(P)$ elementary abelian. A morphism from (P, E) to (R, D) is a morphism $\varphi\colon P \to R$ of $\mathcal{O}(\mathcal{F}^c)$ with $D \leqslant \varphi(E)$. The functor $\pi\colon \mathcal{P} \to \mathcal{O}(\mathcal{F}^c)$ given by $(P, E) \mapsto P$ has the left adjoint $P \mapsto (P, \Omega_1(Z(P)))$. Let $\mathcal{P}_0$ be the full subcategory of $\mathcal{P}$ whose objects are the pairs (P, E) with E in $\mathcal{F}^e$. By the extension property of saturated fusion systems, every pair is isomorphic to an object of $\mathcal{P}_0$, so the inclusion $j\colon \mathcal{P}_0 \to \mathcal{P}$ is an equivalence. Precomposition with π and j preserves higher limits. Thus the functor $F_0 = F \circ \pi^{\mathrm{op}} \circ j^{\mathrm{op}}$ on $\mathcal{P}_0$ has the same higher limits as F.

The functor $\tau\colon \mathcal{P}_0^{\mathrm{op}} \to \mathcal{F}^e$ sends the morphism of $\mathcal{P}_0^{\mathrm{op}}$ given by $\varphi\colon (P, E) \to (R, D)$ to the restriction of φ^{-1} to D. For E in $\mathcal{F}^e$, let $\mathcal{P}_E$ be the subcategory of $\mathcal{P}_0$ whose objects are the pairs (P, E) and whose morphisms are the φ with $\varphi|_E = \mathrm{Id}_E$, and let $j_E\colon \mathcal{P}_E \to \mathcal{P}_0$ be the inclusion. The category $\mathcal{P}_E$ is isomorphic to $\mathcal{O}(C_{\mathcal{F}}(E)^c)$, and the opposite of $\mathcal{P}_E$ is the fibre of τ over E. Let $E \downarrow \tau$ be the undercategory whose objects are the pairs (a, θ) with a in $\mathcal{P}_0$ and $\theta\colon E \to \tau(a)$ a morphism of $\mathcal{F}^e$. The functor $\mathcal{P}_E \to (E \downarrow \tau)^{\mathrm{op}}$ that sends b to (b, Id_E) has a left adjoint. The construction of this left adjoint uses, for each object (a, θ) of $E \downarrow \tau$, a factorisation that comes from the extension property of saturated fusion systems. Let $Y\colon \mathcal{P}_0^{\mathrm{op}} \to A\text{-}\mathtt{mod}$ be a functor. By Lemma 2.3 in degree 0, the value at E of the right Kan extension $\mathrm{Ran}_\tau Y$ of Y along τ [35, X.3, Theorem 1 and Corollary 2] is isomorphic to the inverse limit of $Y \circ j_E^{\mathrm{op}}$ over $\mathcal{P}_E$, naturally in Y.

For functors $X\colon \mathcal{F}^e \to A\text{-}\mathtt{mod}$, precomposition $X \mapsto X \circ \tau$ is exact, so its right adjoint Ran_τ sends injectives to injectives. For functors $Y\colon \mathcal{P}_0^{\mathrm{op}} \to A\text{-}\mathtt{mod}$, precomposition $Y \mapsto Y \circ j_E^{\mathrm{op}}$ has a left adjoint [35, X.3, formulas (9) and (10)] given by a coproduct of evaluations. That adjoint is exact, so precomposition also sends injectives to injectives. For an injective resolution $I^\bullet$ of F_0, the complex $\mathrm{Ran}_\tau I^\bullet$ consists of injectives and is exact by the hypothesis of Lemma 4.1, including in degree 0. It is an injective resolution of the zero functor. Applying the inverse limit and using the isomorphism $\varprojlim_{(\mathcal{F}^e)^{\mathrm{op}}} \mathrm{Ran}_\tau Y \cong \varprojlim_{\mathcal{P}_0} Y$, natural in Y, proves the lemma.

Formalisation.

The principal declarations are `Sharpness.FusionSystem.V28.centralizer_higherLimits_zero_routeB` and `DPV28CentralizerRouteB.centralizer_detection`. Their proofs use the category of pairs, the fibres of τ and the argument with injective resolutions in the proof of Lemma 4.1. In Lean, the higher limits are Ext groups from the constant functor with value A, whose Hom functor is naturally isomorphic to the inverse limit. In Rocq, they are the cohomology of the complex of inverse limits of an injective resolution. The isomorphisms of inverse limits in this argument are natural with respect to morphisms of functors, and so they commute with the differentials of the resolution. The coefficient ring may be any commutative ring in Lean and any ring with the Mathematical Components structure `pzRingType` in Rocq.

4.2 Vanishing of the inverse limit

In Lemma 4.2, $\mathcal{H}$ is a saturated subsystem of $\mathcal{F}$ over $T \leqslant S$, and every $\mathcal{H}$-centric subgroup is $\mathcal{F}$-centric. Let M be a simple Mackey functor for $\mathcal{F}$ over $\mathbb{F}_p$ whose minimal subgroups are not $\mathcal{F}$-centric. A compatible family over $\mathcal{O}(\mathcal{H}^c)$ is determined by its value x at T. As the family is compatible, the restriction of x to each $\mathcal{H}$-centric subgroup P is invariant under $\mathrm{Aut}_{\mathcal{H}}(P)$. By Alperin's fusion theorem [10, Theorem 4.52], the restrictions of x are stable under every isomorphism in $\mathcal{H}$.

Let Q be a minimal subgroup for M and let $L \leqslant T$ be an $\mathcal{F}$-conjugate of Q. Since L is not $\mathcal{F}$-centric and every $\mathcal{H}$-centric subgroup is $\mathcal{F}$-centric, L is not $\mathcal{H}$-centric. Thus $C_T(L_0) \not\leqslant L_0$ for some $\mathcal{H}$-conjugate L_0 of L. By Lemma 2.8(iii), the restriction of x to L_0 is zero. As the restrictions of x are stable under every isomorphism in $\mathcal{H}$, the restriction of x to L also vanishes. This holds for every $\mathcal{F}$-conjugate of Q contained in T, and so $x = 0$ by Lemma 2.8(ii).

Formalisation.

The principal declarations are `Sharpness.FusionSystem.V29.subsystemHigherLimitZero` and `DPV29FusionStableZero.subsystem_centric_limit_zero`. In Lean the conclusion concerns the higher limit in degree 0, and in Rocq the module of compatible families. The isomorphism between this higher limit and the inverse limit relates these formulations. The result is proved in Lean over any field of characteristic p, and in Rocq over a commutative ring in which $p = 0$. Both proofs use the formal counterparts of Lemma 2.8(ii) and (iii) described in Chapter 2.

Alperin's fusion theorem in its form for centric subgroups is proved in both systems. The theorems `Sharpness.FusionSystem.exists_centricFactorization_iso` and `DPFusionAlperinTheorem.centric_factorization_iso` state that each isomorphism of a saturated fusion system factorises, as a group homomorphism, into a finite composite of restrictions of automorphisms of centric subgroups. The Rocq proof of Lemma 4.2 uses this factorisation. For an arbitrary morphism, `Sharpness.FusionSystem.exists_centricFactorization_hom` and `DPFusionAlperinTheorem.centric_factorization_hom` state that it factorises in this way, with the inclusion of the image added at the end. The Lean proof of Lemma 4.2 uses the separately proved form of Alperin's fusion theorem for the orbit category, `Sharpness.FusionSystem.alperinStronglyGenerated_iso`, through `Sharpness.FusionSystem.alperinInput`. In each system saturation consists of the Sylow and extension properties. Alperin's fusion theorem is proved from those properties and is not part of the definition.

4.3 Mackey functors for the centralisers

The hypothesis of Proposition 4.3 is that, for every E in $\mathcal{F}^e$ and every Mackey functor N for $C_{\mathcal{F}}(E)$ over $\mathbb{F}_p$, the higher limits of N^* over $\mathcal{O}(C_{\mathcal{F}}(E)^c)$ vanish in positive degrees. By Lemma 2.9, it suffices to prove that the higher limits of M^* over $\mathcal{O}(\mathcal{F}^c)$ vanish in positive

degrees when M is a simple Mackey functor for $\mathcal{F}$ over $\mathbb{F}_p$ whose minimal subgroups are not $\mathcal{F}$-centric. When $S = 1$ there are no such functors. Suppose that $S \neq 1$ and fix such a functor M. Let E be in $\mathcal{F}^e$. Then the fusion system $C_{\mathcal{F}}(E)$ is saturated and every $C_{\mathcal{F}}(E)$-centric subgroup is $\mathcal{F}$-centric. By Lemma 4.2, the inverse limit of M^* over $\mathcal{O}(C_{\mathcal{F}}(E)^c)$ is zero. The restriction of M to $C_{\mathcal{F}}(E)$ is a Mackey functor for $C_{\mathcal{F}}(E)$ over $\mathbb{F}_p$ whose contravariant part agrees with M^* on $\mathcal{O}(C_{\mathcal{F}}(E)^c)$. By the hypothesis of Proposition 4.3, the higher limits of M^* over $\mathcal{O}(C_{\mathcal{F}}(E)^c)$ also vanish in positive degrees. As E is arbitrary, it follows from Lemma 4.1 that the higher limits of M^* over $\mathcal{O}(\mathcal{F}^c)$ vanish in every degree. By Lemma 2.9, the higher limits over $\mathcal{O}(\mathcal{F}^c)$ of the contravariant part of every Mackey functor for $\mathcal{F}$ over $\mathbb{F}_p$ then vanish in positive degrees.

Formalisation.

The principal declarations are `Sharpness.FusionSystem.V29.centralizer_sharpness_of_local` and `DPV29DiazPark.centralizer_criterion` in `DiazParkV29.MainField`. The Lean statement has coefficients in $\mathbb{F}_p$, while the Rocq statement allows any field of characteristic p. Neither requires $S \neq 1$. Saturation of $C_{\mathcal{F}}(E)$ is proved by `Sharpness.FusionSystem.centralizerSystem_saturated` and `DPFusionCentralizerSaturation.centralizer_saturated`. In Lean, `AKOCentralizerInput` is constructed by `akoCentralizerInput`, and `YalcinCentralizerCentricInput` by `yalcinCentralizerCentricInput`. In these constructions it is proved that $C_{\mathcal{F}}(E)$ is saturated and that every $C_{\mathcal{F}}(E)$-centric subgroup is $\mathcal{F}$-centric. Thus neither fact is an additional hypothesis of the Lean statement. Before applying Lemma 4.1, both proofs show that the restrictions to $\mathcal{O}(C_{\mathcal{F}}(E)^c)$ of M^* and of the contravariant part of the restriction of M to $C_{\mathcal{F}}(E)$ agree. In Lean they are related by an isomorphism whose components are identity maps, and in Rocq they are proved equal.

Chapter 5

The proof of Theorem 1.1 and its consequences

The proof first treats coefficients in $\mathbb{F}_p$, by induction on $|S|$ over all saturated fusion systems $\mathcal{F}$ over finite p-groups S. Over $\mathbb{Z}_{(p)}$, the argument uses the kernel and cokernel of multiplication by p, Nakayama's lemma and Lemma 2.6(ii). Corollary 1.2 on group cohomology then follows from Theorem 1.1. The proofs of the topological consequences of Corollary 1.2 also use geometric realisations and singular homology.

5.1 Induction on the order of the p-group

Let $\mathcal{F}$ be a saturated fusion system over a finite p-group S. The conclusion of Proposition 5.1 is that $\varprojlim^i{}_{\mathcal{O}(\mathcal{F}^c)} M^* = 0$ for every $i > 0$ and every Mackey functor M for $\mathcal{F}$ over $\mathbb{F}_p$. By Proposition 4.3, it suffices to prove this vanishing for the centralisers $C_{\mathcal{F}}(E)$, where E is nontrivial, elementary abelian and fully centralised. Such a centraliser is saturated over $C_S(E)$, and E is central in it. Its quotient by E is saturated by [2, Lemma II.5.5], and $|C_S(E)/E| < |S|$. The higher limits for this quotient vanish in positive degrees by the induction hypothesis, and hence those for $C_{\mathcal{F}}(E)$ by Proposition 3.2. This proves Proposition 5.1, in which the values of the Mackey functor need not be finite dimensional.

Formalisation.

`Sharpness.FusionSystem.V29.primeField_sharpness` and `DiazParkV29.MainField.DPV29DiazPark.field_sharpness` follow this induction. Both quantify over finite p-groups and saturated fusion systems. The Rocq theorem allows every field of characteristic p. In Lean, the induction is carried out with coefficients in $\mathbb{F}_p$, and the case of an arbitrary field of characteristic p is deduced from Theorem 1.1 over $\mathbb{Z}_{(p)}$. The centralisers and the quotients are constructed as fusion systems, and their saturation is proved. In particular, `Sharpness.FusionSystem.akoCentralizerInput`, `Sharpness.FusionSystem.akoQuotientInput` and `Sharpness.FusionSystem.yalcinCentralizerCentricInput` are proved constructions used in this induction. Neither theorem assumes $S \neq 1$.

5.2 Coefficients in $\mathbb{Z}_{(p)}$ and group cohomology

For a Mackey functor M over $\mathbb{Z}_{(p)}$ whose values are finitely generated, let K, pM and M/pM be the kernel, image and cokernel of multiplication by p. Since p annihilates K and M/pM, both are Mackey functors over $\mathbb{F}_p$. By Proposition 5.1, the higher limits of K^* and $(M/pM)^*$

over $\mathcal{O}(\mathcal{F}^c)$ vanish in all positive degrees. These higher limits do not change when the values are regarded as $\mathbb{Z}_{(p)}$-modules. Hence the long exact sequences of higher limits for $0 \to K \to M \to pM \to 0$ and $0 \to pM \to M \to M/pM \to 0$ contain surjections

$$\varprojlim^i_{\mathcal{O}(\mathcal{F}^c)} M^* \longrightarrow \varprojlim^i_{\mathcal{O}(\mathcal{F}^c)} (pM)^* \longrightarrow \varprojlim^i_{\mathcal{O}(\mathcal{F}^c)} M^* \qquad (i > 0).$$

Their composite is multiplication by p. The centric orbit category is finite, and the values of M are finitely generated over the noetherian ring $\mathbb{Z}_{(p)}$. Therefore $\varprojlim^i_{\mathcal{O}(\mathcal{F}^c)} M^*$ is finitely generated by [2, Proposition III.5.5(a)]. By Nakayama's lemma, it is zero. For an arbitrary Mackey functor M, the first case applies to every Mackey subfunctor of M whose values are finitely generated. Since M is the direct limit of these subfunctors, the vanishing for M follows by Lemma 2.6(ii).

A Mackey functor over a field of characteristic p can be regarded as a Mackey functor over $\mathbb{Z}_{(p)}$ with the same higher limits, so Theorem 1.1 also holds over every field of characteristic p. For each $n \geqslant 0$, the cohomology functor $H^n(-;\mathbb{F}_p)$ is the restriction to $\mathcal{O}(\mathcal{F}^c)$ of the contravariant part of a Mackey functor for $\mathcal{F}$ over $\mathbb{F}_p$. Therefore Corollary 1.2 follows from Theorem 1.1 over $\mathbb{F}_p$.

Formalisation.

Theorem 1.1 is `Sharpness.FusionSystem.V29.primeLocal_sharpness` in Lean and `DiazParkV29.MainManuscript.DPV29DiazParkManuscript.prime_local_main` in Rocq. In both proofs, the case of Mackey functors whose values are finitely generated is proved first, and the general case follows by Lemma 2.6(ii). The Lean statement for the form of the Mackey formula for inclusions is `Sharpness.FusionSystem.V29.inclusion_primeLocal_acyclic`. The Rocq statement uses that form directly. In `Sharpness.FusionSystem.fullMackeyOfInclusionIdentity` and `DiazPark.MackeyFusionBridge.DPMackeyFusion.full_mackey_of_inclusions`, the Mackey formula for all morphisms is deduced from the formula for inclusions without changing the given values, restrictions and transfers. The formal statements of Theorem 1.1 have no hypothesis of finite generation.

Corollary 1.2 is `Sharpness.FusionSystem.V29.cohomology_primeField_higherLimit_isZero` in Lean and `DiazParkV29.MainCohomology.DPV29GroupCohomologySharpness.prime_field_higher_limits_zero` in Rocq. The more general declarations `Sharpness.FusionSystem.V29.cohomology_acyclic` and `DiazParkV29.MainCohomology.DPV29GroupCohomologySharpness.field_higher_limits_zero` allow every field k of characteristic p and every k-module with trivial group action. Before these declarations are applied, the restriction and transfer maps of group cohomology are constructed and the Mackey formula for them is proved. The resulting Mackey functor is `Sharpness.FusionSystem.nativeGroupCohomologyMackey` in Lean and `DiazParkV16b.GroupCohomologyMackeyNative.DPGroupCohomologyMackeyNative.native_cohomology_mackey` in Rocq. It is also proved that the restriction to $\mathcal{O}(\mathcal{F}^c)$ of the contravariant part of this Mackey functor is the cohomology functor, up to an isomorphism in Lean and as an equality in Rocq.

5.3 Subgroup and normaliser decompositions

The abstract of the manuscript states that the subgroup and normaliser decompositions of every p-local finite group $(S, \mathcal{F}, \mathcal{L})$ over the $\mathcal{F}$-centric subgroups are sharp. Both decompositions are described in Section 1 of the manuscript, where the sharpness of the subgroup decomposition is deduced from Corollary 1.2. For a p-local finite group $(S, \mathcal{F}, \mathcal{L})$, the subgroup decomposition is indexed by $\mathcal{O}(\mathcal{F}^c)$. Its Bousfield–Kan spectral sequence has $E_2^{i,n} = \varprojlim^i_{\mathcal{O}(\mathcal{F}^c)} H^n(-;\mathbb{F}_p)$. By

Corollary 1.2, $E_2^{i,n} = 0$ for all $i > 0$ and $n \geqslant 0$. As recalled in Section 1 of the manuscript, Yalçın [53, Theorem 1.9] proved that the subgroup decomposition is sharp if and only if the normaliser decomposition of Libman [34] over the $\mathcal{F}$-centric subgroups is sharp.

Formalisation.

In Lean, the theorems `subgroup_decomposition_sharp` and `normalizer_decomposition_sharp`, in the namespace `Sharpness.FusionSystem.CentricLinkingSystem.V29`, state that, for every $n \geqslant 0$, the higher limits of the composite of the functor of each decomposition with mod p singular cohomology in degree n vanish in positive degrees. Both theorems also assert that the restriction map to the inverse limit is invertible. In Rocq, `DiazParkV29.MainCollapse.DPV29LinkingSharpnessCollapse.subgroup_collapses_onto_vertical_axis` and `DiazParkV29.MainCollapse.DPV29LinkingSharpnessCollapse.normalizer_collapses_onto_vertical_axis` state that the constructed spectral sequences of filtered cochain complexes collapse onto the vertical axis. On every page from E_2 on, their terms in positive columns and all their differentials vanish. In both systems, the proofs of these statements use the vanishing for the cohomology functors proved in `Sharpness.FusionSystem.V29.cohomology_acyclic` and `DiazParkV29.MainCohomology.DPV29GroupCohomologySharpness.field_higher_limits_zero`. For the formal results on the normaliser decomposition, this vanishing is also applied to the saturated normaliser fusion systems of chains.

The formal results on both decompositions have a centric linking system associated to $\mathcal{F}$ among their hypotheses, but no decomposition or sharpness theorem. They hold for every field k of characteristic p and every k-module with trivial group action. The Rocq statements that involve geometric realisation are quantified over `Real : realType`, which serves as the real numbers in that construction. The Lean formalisation uses Mathlib's geometric realisation. The proofs of the formal results on the normaliser decomposition rest on the proved stable elements theorem for finite groups with trivial coefficients (see, for example, [R1, Proposition III.10.4]). Excision for singular homology, for a cover by two open subspaces, enters the proofs for both decompositions, and so does the isomorphism between the simplicial homology of a simplicial set and the singular homology of its geometric realisation. In Lean, the stable elements theorem for finite groups, excision and this isomorphism are `Sharpness.OrdinaryCohomology.restrictionStableElementsEquiv`, `Sharpness.SingularChainExcision.relativeExcisionMap_quasiIso` and `Sharpness.realizationUnitChainMap_quasiIso_unconditional`, and in Rocq they are `DiazParkV16b.GroupCohomologyStableElementsNative.DPGroupCohomologyStableElementsNative.finite_group_stable_iso`, `DiazParkV16b.SingularExcisionQuasiIso.DPSingularExcisionQuasiIso.relative_excision_quasi_isomorphism` and `DiazParkV16b.RealizationUnitComparison.DPRealizationUnitComparison.realization_unit_quasi_isomorphism`. Thus the formal proofs of these topological consequences use results of topology in addition to the vanishing for the cohomology functors.

5.4 The stable elements theorem

The stable elements theorem of Broto, Levi and Oliver [6, Theorem 5.8] is cited in Section 1 of the manuscript. As recalled in Section 2.1 of the manuscript, Henke, Libman and Lynd [27, §1.1] noted that this theorem is a direct consequence of the sharpness of the subgroup decomposition. In both formalisations, the following isomorphism is proved

$$H^n(|\mathcal{L}|; \mathbb{F}_p) \longrightarrow \varprojlim_{\mathcal{O}(\mathcal{F}^c)} H^n(-; \mathbb{F}_p).$$

Here $|\mathcal{L}|$ is the geometric realisation of the nerve of $\mathcal{L}$, without p-completion. For each $\mathcal{F}$-centric subgroup P, let BP be the classifying space of P, the geometric realisation of the nerve of P regarded as a category with one object. The distinguished monomorphism from P to $\mathrm{Aut}_{\mathcal{L}}(P)$ induces a map $BP \to |\mathcal{L}|$. The component at P of this isomorphism is the map induced in

cohomology by $BP \to |\mathcal{L}|$, followed by the isomorphism $H^n(BP;\mathbb{F}_p) \cong H^n(P;\mathbb{F}_p)$ between the singular cohomology of BP and the group cohomology of P.

Formalisation.

In Lean, the restriction map from $H^n(|\mathcal{L}|;\mathbb{F}_p)$ to the inverse limit is `Sharpness.FusionSystem.CentricLinkingSystem.nativeOrdinaryRestrictionToLimit`. In the namespace `Sharpness.FusionSystem.CentricLinkingSystem.V29`, its invertibility is proved in `nativeOrdinaryRestrictionToLimit_isIso`, the resulting isomorphism is `nativeOrdinaryRestrictionIso`, and the formula for its components is proved in `nativeOrdinaryRestrictionIso_component`. The corresponding Rocq declarations are `DiazParkV29.MainSubgroup.DPV29LinkingSubgroup.native_edge_iso` and `DiazParkV29.MainSubgroup.DPV29LinkingSubgroup.native_edge_component`. In these declarations, this restriction map is the restriction in singular cohomology, followed by the map induced on inverse limits by the natural isomorphism to group cohomology.

It is proved in `Sharpness.realizationUnitChainMap_quasiIso_unconditional` and `DiazParkV16b.RealizationUnitComparison.DPRealizationUnitComparison.realization_unit_quasi_isomorphism` that, for every simplicial set, the chain map from its simplicial chains to the singular chains of its geometric realisation is a quasi-isomorphism. This Rocq theorem is used to instantiate the hypothesis `hunit` of the declarations of `LinkingSubgroupRealizationComparison` that enter the construction of the isomorphism and the proof of the formula for its components. The augmentation of the subgroup decomposition is the cochain map from the singular cochains of $|\mathcal{L}|$ to the total complex of a double complex. The term of this double complex in bidegree (i,q) is the product, over all composable sequences $P_0 \to \cdots \to P_i$ of i morphisms in $\mathcal{O}(\mathcal{F}^c)$, of the singular q-cochains on the space that the functor of the subgroup decomposition assigns to P_0. In Lean, this augmentation is proved to be a quasi-isomorphism in `Sharpness.FusionSystem.CentricLinkingSystem.subgroupCochainAugmentation_quasiIso_unconditional`. The Lean and Rocq isomorphisms from $H^n(|\mathcal{L}|;\mathbb{F}_p)$ to the inverse limit are constructed for all $n \geqslant 0$ and all finite p-groups, including $S = 1$. More generally, coefficients in every k-module with trivial group action are allowed, for every field k of characteristic p.

Appendix A

Manuscript statements and formal declarations

For every numbered statement, including definitions, a principal declaration in each system is named in Table A.1, and the relation of these declarations to the statement is described in the last column. The hypotheses, coefficients and degrees of each declaration are given by its formal statement.

Each entry on use concerns only the declaration in its row and the formal proofs of Theorem 1.1 named in the first row.

Table A.1: Manuscript statements and formal declarations.

Manuscript	Formal declarations	Relation to the statement
Theorem 1.1	Lean `Sharpness.FusionSystem.V29.primeLocal_sharpness` Rocq `DiazParkV29.MainManuscript.DPV29DiazParkManuscript.prime_local_main`	Whole statement.
Corollary 1.2	Lean `Sharpness.FusionSystem.V29.cohomology_primeField_higherLimit_isZero` Rocq `DiazParkV29.MainCohomology.DPV29GroupCohomologySharpness.prime_field_higher_limits_zero`	Whole statement. Not used in Lean.
Definition 2.1	Lean `Sharpness.FusionSystem.Orbit` Rocq `DiazPark.FusionOrbit.DPFusionOrbit.orbit_category`	Definition. Used in both proofs. The declarations define the orbit category.
Definition 2.2	Lean `Sharpness.FusionSystem.CentralSubgroup` Rocq `DiazPark.FusionQuotientMaps.DPFusionQuotient.central`	Definition. Used in Lean. The formal definition of central subgroups is proved equivalent to that of Definition 2.2.
Lemma 2.3	Lean `Sharpness.Unconditional.higherLimitRightAdjointNatIso` Rocq `DiazPark.CategoryDerivedAdjunction.derived_left_adjoint_nat_iso`	Whole statement. Used in Rocq. Not used in Lean. The adjunction is taken on opposite categories.

Manuscript	Formal declarations	Relation to the statement
Definition 2.4	Lean `Sharpness.MackeyModule` Rocq `DiazPark.MackeyStructures.mackey_module`	Definition. Used in Lean. The Mackey formula for inclusions is proved equivalent to its form for all morphisms of $\mathcal{F}$.
Definition 2.5	Lean `Sharpness.MackeyModule.Subfunctor` Rocq `DiazPark.MackeyStructures.mackey_subfunctor`	Definition. Used in Lean. The declarations define Mackey subfunctors.
Lemma 2.6(i)	Lean `Sharpness.MackeyModule.finiteGeneratedSubfunctor` Rocq `DiazPark.MackeyFusionFiniteGeneration.DPMackeyFinite.generated_values_finitely_generated`	A part of the proof. Used in both proofs. The construction of the Mackey subfunctor generated by a finite family of elements and the finite generation of its values have separate declarations.
Lemma 2.6(ii)	Lean `Sharpness.FusionSystem.higherLimit_isZero_of_fg_subfunctors_v29` Rocq `DiazParkV29.CompositionFilteredMackey.DPFilteredLimitsMackey.centric_subfunctor_higher_limit_closure`	Whole statement. Used in both proofs. A fixed degree may be 0.
Definition 2.7	Lean `Sharpness.FusionSystem.IsMinimalSupport` Rocq `DiazPark.MackeyFusionMinimal.DPMackeyMinimal.subgroup_minimal`	Definition. Used in both proofs. The Lean definition of minimal subgroups is stated over a field.
Lemma 2.8(i)	Lean `Sharpness.FusionSystem.simple_minimalSupport_class` Rocq `DiazParkV16b.MackeyMinimalSupportManuscript.DPMackeyMinimalManuscript.minimal_support_iff_conjugate`	A part of the proof. Used in Lean. In Rocq, the simplicity of the modules at the minimal subgroups has a separate declaration.
Lemma 2.8(ii)	Lean `Sharpness.FusionSystem.simple_functor_restriction_detection_v29` Rocq `DiazParkV29.SimpleFunctor.DPSimpleFunctor.simple_functor_restriction_detection`	Whole statement. Used in both proofs.
Lemma 2.8(iii)	Lean `Sharpness.FusionSystem.simple_centralizer_zero_v29` Rocq `DiazParkV29.SimpleManuscript.DPSimpleManuscript.simple_functor_centralizer_restriction_zero`	Whole statement. Used in both proofs. Saturation is not a hypothesis of these declarations.
Lemma 2.9	Lean `Sharpness.FusionSystem.V29.noncentric_simple_reduction` Rocq `DiazParkV29.ReductionManuscript.DPV29MackeyReduction.noncentric_simple_to_arbitrary_higher_limits`	Whole statement. Used in both proofs.
Lemma 3.1(i)	Lean `Sharpness.FusionSystem.quotientCollection_centric` Rocq `DiazPark.FusionQuotientCollection.DPFusionQuotientCollection.collection_centric`	A part of the proof. Used in both proofs. The closure properties and the inclusion $\mathcal{F}^{cr} \subseteq \mathcal{X}$ have separate declarations.

Manuscript	Formal declarations	Relation to the statement
Lemma 3.1(ii)	Lean `Sharpness.FusionSystem.quotientActualCentricIsoCatUnconditional` Rocq `DiazParkV16b.FusionSharpnessManuscript.DPDiazParkManuscript.quotient_collection_category_isomorphism`	Whole statement. Not used in Lean. The formulas of the isomorphism on objects and morphisms are proved separately.
Proposition 3.2	Lean `Sharpness.FusionSystem.V29.central_quotient_sharpness` Rocq `DiazParkV29.MainCentralQuotient.DPV29CentralQuotientSharpness.central_quotient_sharpness`	Whole statement. Used in both proofs.
Lemma 4.1	Lean `Sharpness.FusionSystem.V28.centralizer_higherLimits_zero_routeB` Rocq `DiazParkV28.FusionCentralizerRouteB.DPV28CentralizerRouteB.centralizer_detection`	Whole statement. Used in both proofs. The formal proofs follow the algebraic proof of the manuscript.
Lemma 4.2	Lean `Sharpness.FusionSystem.V29.subsystemHigherLimitZero` Rocq `DiazParkV29.SubsystemDegreeZero.DPV29FusionStableZero.subsystem_centric_limit_zero`	Whole statement. Used in both proofs.
Proposition 4.3	Lean `Sharpness.FusionSystem.V29.centralizer_sharpness_of_local` Rocq `DiazParkV29.MainField.DPV29DiazPark.centralizer_criterion`	Whole statement. Used in both proofs.
Proposition 5.1	Lean `Sharpness.FusionSystem.V29.primeField_sharpness` Rocq `DiazParkV29.MainField.DPV29DiazPark.field_sharpness`	Whole statement. Used in both proofs.

Appendix B

Sources used in the manuscript

This index groups citations by reference and locator. The second column locates each occurrence in the manuscript. The counts describe occurrences only, and they neither confirm how a cited result is applied nor mean that it is proved in Lean or Rocq.

Table B.1 covers the abstract and Sections 1–5. It contains 117 occurrences of bibliography keys in 115 citation commands. The abstract has no citation command, and the three works written out in brackets there are neither counted nor located in the table. A command citing two works contributes one occurrence for each key. The report bibliography contains entries 1–53 of the manuscript bibliography, in the same order and with the same numbers.

Table B.1: Sources cited in Sections 1–5 of the manuscript.

Reference and locator	Location in the manuscript	Occurrences
[1]	Section 1, discussion	1
[2]	Section 2, discussion	1
[2, Definition II.5.3]	Section 3, discussion	1
[2, Lemma II.5.5]	Section 3, discussion Section 5, proof of Proposition 5.1	2
[2, Proposition III.5.5(a)]	Section 5, proof of Theorem 1.1	1
[2, Proposition III.5.5(b)]	Section 2.3, proof of Lemma 2.6	1
[2, §III.5.1]	Section 2.2, discussion	1
[2, §III.7]	Section 1, discussion	2
[2, §III.7, Problem 1]	Section 1, discussion	1
[3]	Section 1, discussion	1
[4]	Section 3, Lemma 3.1	1
[4, Corollary 6.14]	Section 3, proof of Lemma 3.1	1
[4, Corollary 6.14, p. 3856]	Section 3, proof of Lemma 3.1	1
[4, Lemma 6.4(a)]	Section 3, proof of Lemma 3.1	1
[4, p. 3856, equation (3)]	Section 3, proof of Lemma 3.1	1
[5, Lemma 1.3]	Section 1, discussion	1
[6]	Section 1, discussion	1
[6, Definition 2.4 and Proposition 2.5(c)]	Section 2.1, discussion	1

Reference and locator	Location in the manuscript	Occurrences
[6, Proposition 2.2]	Section 1, discussion	1
[6, Proposition 2.5(a)]	Section 2.2, discussion Section 4, proof of Lemma 4.1	2
[6, Proposition 3.1]	Section 1, discussion	1
[6, Theorem 2.6]	Section 2.1, discussion	1
[6, Theorem 5.8]	Section 1, discussion	1
[6, §9]	Section 1, discussion	1
[6, proof of Corollary 3.6]	Section 3, proof of Proposition 3.2	1
[6, proof of Theorem 2.6]	Section 4, discussion	1
[6, proof of Theorem 5.8]	Section 2.1, discussion	2
[7, Conjecture 57.12]	Section 1, discussion	1
[8]	Section 1, discussion	1
[9]	Section 1, discussion	2
[10, Theorem 4.52]	Section 4, proof of Lemma 4.2	1
[11]	Section 1, discussion	1
[12, Proposition 2.9 and Remark 2.10]	Section 1, discussion	1
[13]	Section 1, discussion	1
[13, Conjecture]	Section 1, discussion	1
[13, Corollary 3.4]	Section 2.3, proof of Lemma 2.8	1
[13, Definition 2.1]	Section 2.3, Definition 2.4	1
[13, Example 5.4]	Section 1, discussion	1
[13, Proposition 3.1]	Section 2.3, proof of Lemma 2.8	1
[13, Proposition 3.2]	Section 2.3, proof of Lemma 2.8	1
[13, Proposition 4.3]	Section 2.1, discussion Section 2.3, proof of Lemma 2.9	2
[13, Theorem A]	Section 1, discussion	1
[13, Theorem B]	Section 1, discussion	1
[13, Theorem C]	Section 1, discussion	1
[13, §3]	Section 2.3, proof of Lemma 2.8	1
[13, formula (4)]	Section 2.3, proof of Lemma 2.8	1
[13, formula (6)]	Section 2.3, proof of Lemma 2.8	1
[14]	Section 1, discussion	1
[15, Theorem 10.3]	Section 1, discussion	1
[15, §1]	Section 1, discussion	1
[16, Theorem 1.2]	Section 2.1, discussion	1
[17, Appendix II, Proposition 3.3]	Section 2.2, discussion	1
[18]	Section 1, discussion	1
[19]	Section 1, discussion	1
[20, Corollary 5.3.3]	Section 3, proof of Proposition 3.2	1
[21, Lemma 2.9]	Section 2.3, proof of Lemma 2.8	1
[21, Theorem A]	Section 1, discussion	1

Reference and locator	Location in the manuscript	Occurrences
[22]	Section 1, discussion	1
[23]	Section 1, discussion	1
[24, Theorem 1.2]	Section 1, discussion	1
[24, Theorem 1.5]	Section 1, discussion	1
[25, Theorem 1.1 and Remark 3.2]	Section 1, discussion	1
[26, Theorem A]	Section 1, discussion	1
[27, Definition 2.16]	Section 1, discussion	1
[27, Theorem 1.1]	Section 1, discussion	1
[27, Theorem 1.4]	Section 1, discussion	1
[27, Theorem 5.2(c),(d)]	Section 1, discussion	1
[27, §1.1]	Section 2.1, discussion	1
[28]	Section 1, discussion	1
[29]	Section 1, discussion	1
[30, Corollary 5.16]	Section 1, discussion	2
[30, Lemma 3.1]	Section 2.2, Lemma 2.3	1
[30, Proposition 5.14]	Section 1, discussion	1
[31, Proposition 5.1(ii)]	Section 2.2, discussion	1
[32]	Section 2.1, discussion	1
[32, Theorem 0.6]	Section 2.1, discussion	1
[33]	Section 1, discussion	1
[34]	Section 1, discussion	2
[35, X.3, Theorem 1 and Corollary 2]	Section 4, proof of Lemma 4.1	1
[35, X.3, formulas (9) and (10)]	Section 4, proof of Lemma 4.1	1
[36]	Section 1, discussion	1
[37]	Section 1, discussion	1
[38]	Section 1, discussion	1
[39, Theorem A(c)]	Section 1, discussion	1
[40]	Section 1, discussion	1
[41]	Section 1, discussion	1
[42]	Section 1, discussion	2
[43]	Section 1, discussion	1
[44, Theorem C]	Section 1, discussion	1
[44, §4]	Section 1, discussion	1
[45]	Section 1, discussion	1
[45, Theorem D]	Section 1, discussion	1
[45, §4]	Section 1, discussion	1
[46]	Section 1, discussion	1
[47]	Section 1, discussion	1
[48]	Section 1, discussion	1
[48, §1.3]	Section 1, discussion	1
[49]	Section 1, discussion	1
[50]	Section 1, discussion	1
[51]	Section 1, discussion	1
[52, Proposition 2.1]	Section 2.3, proof of Lemma 2.8	1
[52, Proposition 2.8]	Section 2.3, proof of Lemma 2.8	1
[53, Conjecture 1.2]	Section 1, discussion	1
[53, Lemma 9.3 and Theorem 9.4]	Section 4, discussion	1
[53, Theorem 1.8]	Section 1, discussion Section 2.1, discussion	2
[53, Theorem 1.9]	Section 1, discussion	1

Reference and locator	Location in the manuscript	Occurrences
[53, §9, p. 55]	Section 2.1, discussion	1

Appendix C

Further formal results

C.1 The sharpness of the centraliser decomposition

The results of this section belong to the supplement on the centraliser decomposition. For a p-local finite group $(S, \mathcal{F}, \mathcal{L})$ with $S \neq 1$, Broto, Levi and Oliver [6, proof of Theorem 5.8] proved that the centraliser decomposition is sharp, as a step in the proof of the stable elements theorem. By an observation of Yalçın [53, §9, p. 55], the centraliser decomposition is sharp if the subgroup decomposition of $(S, \mathcal{F}, \mathcal{L})$ is sharp and, for each nontrivial elementary abelian subgroup E of S fully centralised in $\mathcal{F}$, the subgroup decomposition of the p-local finite group $(C_S(E), C_{\mathcal{F}}(E), C_{\mathcal{L}}(E))$ is sharp, where $C_{\mathcal{L}}(E)$ is the centric linking system that Broto, Levi and Oliver [6, Definition 2.4 and Proposition 2.5(c)] constructed from $\mathcal{L}$. In Section 2.1 of the manuscript, the sharpness of the centraliser decomposition follows from Corollary 1.2 and this observation. In the supplement, this sharpness is proved in Lean and Rocq using Corollary 1.2 and the stable elements theorem.

Let p be a prime, let $S \neq 1$ be a finite p-group, let $\mathcal{F}$ be a saturated fusion system over S and let $\mathcal{L}$ be a centric linking system associated to $\mathcal{F}$. Write $\mathcal{F}^e$ for the full subcategory of $\mathcal{F}$ whose objects are the nontrivial elementary abelian subgroups of S that are fully centralised in $\mathcal{F}$. For E in $\mathcal{F}^e$, let $\overline{C}_{\mathcal{L}}(E)$ be the category defined in the statement of the centraliser decomposition. The objects of this category are the pairs (P, α), where P is $\mathcal{F}$-centric and $\alpha \in \mathrm{Hom}_{\mathcal{F}}(E, Z(P))$. Its morphisms $(P, \alpha) \to (R, \beta)$ are the morphisms $\varphi\colon P \to R$ in $\mathcal{L}$ whose underlying homomorphism, composed with α, is β. For a morphism $E \to D$ in $\mathcal{F}^e$, precomposition gives a functor $\overline{C}_{\mathcal{L}}(D) \to \overline{C}_{\mathcal{L}}(E)$. Thus the nerves of these categories and their geometric realisations form contravariant functors on $\mathcal{F}^e$. The forgetful functors $\overline{C}_{\mathcal{L}}(E) \to \mathcal{L}$, which send (P, α) to P and each morphism to itself, give compatible maps $|\overline{C}_{\mathcal{L}}(E)| \to |\mathcal{L}|$.

For every $n \geqslant 0$, singular cohomology gives the covariant functor

$$E \longmapsto H^n(|\overline{C}_{\mathcal{L}}(E)|; \mathbb{F}_p).$$

Its maps are induced by the maps between these geometric realisations. In the manuscript, higher limits over a category $\mathcal{C}$ are defined for contravariant functors on $\mathcal{C}$. Hence the higher limits of this covariant functor on $\mathcal{F}^e$ are taken over $(\mathcal{F}^e)^{\mathrm{op}}$. In the supplement, it is proved that

$$\varprojlim^{s}_{(\mathcal{F}^e)^{\mathrm{op}}} H^n(|\overline{C}_{\mathcal{L}}(-)|; \mathbb{F}_p) = 0 \qquad (s > 0)$$

and that the canonical map

$$H^n(|\mathcal{L}|; \mathbb{F}_p) \longrightarrow \varprojlim_{(\mathcal{F}^e)^{\mathrm{op}}} H^n(|\overline{C}_{\mathcal{L}}(-)|; \mathbb{F}_p)$$

is an isomorphism for every $n \geqslant 0$. Its component at E is the map induced by $|\overline{C}_{\mathcal{L}}(E)| \to |\mathcal{L}|$. For $s = 0$, the conclusion is not that the higher limit vanishes but that the canonical map is an isomorphism.

Singular cochains with coefficients in $\mathbb{F}_p$ form a covariant functor from $\mathcal{F}^e$ to cochain complexes. It sends E to the singular cochain complex of $|\overline{C}_{\mathcal{L}}(E)|$, and its maps are induced by the maps between these geometric realisations. This functor determines a double complex. Its term in bidegree (i, q) is the product, over all composable sequences $E_0 \to \cdots \to E_i$ of i morphisms in $(\mathcal{F}^e)^{\mathrm{op}}$, of the singular q-cochains on $|\overline{C}_{\mathcal{L}}(E_0)|$. For each $q \geqslant 0$, the terms of bidegree (i, q) and the horizontal differential form the cochain complex induced by the bar resolution for $(\mathcal{F}^e)^{\mathrm{op}}$ with coefficients in the functor that sends E to the singular q-cochains on $|\overline{C}_{\mathcal{L}}(E)|$. The vertical differential is the singular coboundary. The compatible maps $|\overline{C}_{\mathcal{L}}(E)| \to |\mathcal{L}|$ induce a cochain map from the singular cochain complex of $|\mathcal{L}|$ to the total complex of this double complex. This cochain map is the augmentation of the centraliser decomposition. For each E in $\mathcal{F}^e$, the composite of this augmentation with the projection onto the factor indexed by E in the column $i = 0$ is the map of singular cochains induced by $|\overline{C}_{\mathcal{L}}(E)| \to |\mathcal{L}|$. In each degree n, the canonical map is the composite of two maps. The first is the map induced in cohomology by this augmentation. The second is the map from the nth cohomology of the total complex to $\varprojlim_{(\mathcal{F}^e)^{\mathrm{op}}} H^n(|\overline{C}_{\mathcal{L}}(-)|; \mathbb{F}_p)$ given by the projections onto the factors of the column $i = 0$. In the supplement, it is proved that this augmentation is a quasi-isomorphism. Table C.1 lists the principal declarations.

Formalisation.

The Lean declarations of this table lie in `Sharpness.FusionSystem.CentricLinkingSystem`. The Rocq declarations lie in the modules `DPBLOCentralizerEquivalence`, `DPBLOCohomology`, `DPBLOE3Native` and `DPBLOEdgeFinal`, one for each row of the table. Their files are `BLODiagramCentralizerEquivalence`, `BLOCohomology`, `BLOE3Native` and `BLOEdgeFinal` under the Rocq library `DiazParkBLO`. The statements that the augmentation of the centraliser decomposition is a quasi-isomorphism are `BLOEdgeCochainAugmentation_quasiIso` in Lean and `DPBLOEdgeFinal.augmentation_cochain_quasi_isomorphism` in Rocq.

Table C.1: The centraliser decomposition and its formal counterparts.

Statement	Formal declarations	In the manuscript
Centric linking systems of the centralisers and the diagram	Lean `Sharpness.FusionSystem.CentricLinkingSystem.BLODiagramCentralizerEquivalence`. Rocq `DiazParkBLO.BLODiagramCentralizerEquivalence.DPBLOCentralizerEquivalence.centralizer_adjunction`	The diagram of Broto, Levi and Oliver is described in Section 2.1.
Natural isomorphism with a right Kan extension	Lean `Sharpness.FusionSystem.CentricLinkingSystem.BLOCohomologyRightKanIso`. Rocq `DiazParkBLO.BLOCohomology.DPBLOCohomology.native_ran_nat_iso`	The isomorphism is proved in the supplement for the sharpness deduced in Section 2.1.

Statement	Formal declarations	In the manuscript
Positive higher limits	Lean `Sharpness.FusionSystem.CentricLinkingSystem.BLOCohomologyPositiveIsZero`. Rocq `DiazParkBLO.BLOE3Native.DPBLOE3Native.native_positive_higher_limits_zero`	Sharpness is deduced in Section 2.1 from Corollary 1.2 and Yalçın's observation.
Canonical restriction and augmentation	Lean `Sharpness.FusionSystem.CentricLinkingSystem.BLO_centralizer_decomposition_sharp`. Rocq `DiazParkBLO.BLOEdgeFinal.DPBLOEdgeFinal.centralizer_cohomological_sharpness`	Sharpness for $S \neq 1$ is deduced in Section 2.1. In the supplement, it is proved that the canonical restriction is an isomorphism and that the cochain augmentation is a quasi-isomorphism.

The centric linking system $C_{\mathcal{L}}(E)$ of $C_{\mathcal{F}}(E)$ is constructed from $\mathcal{L}$ and proved equivalent to $\overline{C}_{\mathcal{L}}(E)$. Broto, Levi and Oliver [6, Definition 2.4, Proposition 2.5 and Theorem 2.6] used these two categories to describe the centraliser decomposition. Applying the stable elements theorem to this centric linking system gives a natural isomorphism from the functor $E \mapsto H^n(|\overline{C}_{\mathcal{L}}(E)|;\mathbb{F}_p)$ to the right Kan extension along $\tau\colon \mathcal{P}_0^{\mathrm{op}} \to \mathcal{F}^e$ of the functor $(P,E) \mapsto H^n(P;\mathbb{F}_p)$. Here, as in the proof of Lemma 4.1, $\mathcal{P}_0$ is the category of pairs (P,E) of an $\mathcal{F}$-centric subgroup P and a subgroup $E \leqslant Z(P)$ in $\mathcal{F}^e$, and τ sends (P,E) to E. The distinguished monomorphisms $P \to \mathrm{Aut}_{\mathcal{L}}(P)$ of $\mathcal{L}$ determine this isomorphism, and its compatibility with every morphism of $\mathcal{F}^e$ is proved.

By Corollary 1.2, applied to the saturated fusion system $C_{\mathcal{F}}(E)$ for each E in $\mathcal{F}^e$, the higher limits of $H^n(-;\mathbb{F}_p)$ over $\mathcal{O}(C_{\mathcal{F}}(E)^c)$ vanish in positive degrees for every $n \geqslant 0$. This vanishing is needed for the isomorphism, obtained as in the proof of Lemma 4.1, between the higher limits of the right Kan extension over $(\mathcal{F}^e)^{\mathrm{op}}$ and those of $H^n(-;\mathbb{F}_p)$ over $\mathcal{O}(\mathcal{F}^c)$. Applying the corollary to $\mathcal{F}$ then gives $\varprojlim^s{}_{(\mathcal{F}^e)^{\mathrm{op}}} H^n(|\overline{C}_{\mathcal{L}}(-)|;\mathbb{F}_p) = 0$ for every $s > 0$. This isomorphism of higher limits also holds in degree 0. By a commutative square with the restriction map of the stable elements theorem, the canonical map is an isomorphism. The Lean and Rocq proofs that the augmentation of the centraliser decomposition is a quasi-isomorphism show that, for every $n \geqslant 0$, the projections onto the factors of the column $i = 0$ give an isomorphism from the nth cohomology of the total complex to $\varprojlim_{(\mathcal{F}^e)^{\mathrm{op}}} H^n(|\overline{C}_{\mathcal{L}}(-)|;\mathbb{F}_p)$. For each $q \geqslant 0$, the vertical cohomology of the double complex in degree q is isomorphic to the cochain complex induced by the bar resolution for $(\mathcal{F}^e)^{\mathrm{op}}$ with coefficients in the functor $E \mapsto H^q(|\overline{C}_{\mathcal{L}}(E)|;\mathbb{F}_p)$. Since the higher limits of the functor vanish in positive degrees, the cochain complex is exact in these degrees. With this exactness, for each $m \geqslant 1$, every cocycle of the total complex that is zero in the columns $i < m$ is the coboundary of a cochain that is zero in the columns $i < m-1$. This is proved by an induction over the columns $i \geqslant m$, which ends because a cochain of total degree n has components only in the $n+1$ bidegrees $(i, n-i)$ with $0 \leqslant i \leqslant n$. Injectivity of the map given by the projections rests on the case $m = 1$, and surjectivity on the case $m = 2$. The whole Rocq proof also depends on constructed spectral sequences of filtered cochain complexes and on their proved convergence and collapse. It reuses the isomorphism of the stable elements theorem and the formula for its components from the Rocq formalisation, and

this isomorphism is obtained from the collapse of such a spectral sequence for the subgroup decomposition.

The conclusion concerns the cohomology of geometric realisations without p-completion. It does not assert the homotopy equivalence between $|\mathcal{L}|$ and the homotopy colimit in [6, Theorem 2.6], a theorem about p-completion, or compatibility with products or Steenrod operations. The centric linking system $\mathcal{L}$ is a hypothesis, and a construction of a centric linking system for every saturated fusion system is not among the conclusions. In Rocq, the statements that involve geometric realisation are quantified over `Real : realType`, a real closed field with the Archimedean property and the least upper bound property. This field is part of the construction of spaces, rather than a sharpness hypothesis. The case $S = 1$ is discussed in Section 3.2.1, item (ii).

C.2 Other formal proofs of manuscript statements

The formalisations contain other formal proofs of ten manuscript statements. The following Lean declarations are not used by the formal proofs of Theorem 1.1, Corollary 1.2 and the theorems on the sharpness of the subgroup and normaliser decompositions. A Rocq declaration is listed beside each of them. For these Rocq declarations, the saved Rocq dependency sets do not establish non-use.

- Lemma 2.6(ii). Lean `Sharpness.FusionSystem.higherLimit_isZero_of_fg_subfunctors`. Further Rocq declaration `DiazParkV16b.MackeyFixedDegreeClosure.DPMackeyFixedDegree.finite_subfunctor_higher_limit_closure`.
- Lemma 2.8(ii). Lean `Sharpness.FusionSystem.simple_minimal_restriction_detection`. Further Rocq declaration `DiazParkV16b.MackeyMinimalSupportManuscript.DPMackeyMinimalManuscript.minimal_inclusion_restriction_detection`.
- Lemma 2.9. Lean `Sharpness.FusionSystem.noncentric_simple_reduction`. Further Rocq declaration `DiazParkV16b.MackeyNoncentricSimpleReduction.DPMackeyNoncentricReduction.prime_field_noncentric_simple_reduction`.
- Lemma 2.8(iii). Lean `Sharpness.FusionSystem.simple_minimal_centralizer_zero`. Further Rocq declaration `DiazParkV16b.MackeyMinimalSupportManuscript.DPMackeyMinimalManuscript.minimal_centralizer_restriction_zero`.
- Proposition 3.2. Lean `Sharpness.FusionSystem.central_quotient_sharpness_unconditional`. Further Rocq declaration `DiazPark.FusionCentralQuotientSharpness.DPFusionCentralQuotientSharpness.central_quotient_sharpness`.
- Lemma 4.2. Lean `Sharpness.Revised.Baseline.subsystemLimitZero`. Further Rocq declaration `DiazParkV16b.FusionSharpnessManuscript.DPDiazParkManuscript.minimal_subsystem_centric_limit_zero`.
- Proposition 4.3. Lean `Sharpness.FusionSystem.V28.centralizer_sharpness_of_local`. Further Rocq declaration `DiazParkV28.FusionSharpnessFieldRouteB.DPV28DiazPark.centralizer_criterion`.
- Proposition 5.1. Lean `Sharpness.FusionSystem.V28.primeField_sharpness`. Further Rocq declaration `DiazParkV28.FusionSharpnessFieldRouteB.DPV28DiazPark.field_sharpness`.

- Theorem 1.1. Lean `Sharpness.FusionSystem.V28.primeLocal_sharpness`. Further Rocq declaration `DiazParkV28.FusionSharpnessManuscriptRouteB.DPV28DiazParkManuscript.prime_local_main_positive`.
- Corollary 1.2. Lean `Sharpness.FusionSystem.V28.cohomology_primeField_acyclic`. Further Rocq declaration `DiazParkV28.GroupCohomologyRouteB.DPV28GroupCohomologySharpness.prime_field_higher_limits_zero`.

Formalisation.

That a listed Lean declaration is not used does not exclude that the formal proofs of Theorem 1.1 use declarations on which the proof of that declaration rests.

In the other proof of Lemma 2.6(ii), the finitely many components of a cocycle lie in one Mackey subfunctor whose values are finitely generated. The case of degree 0 is included. The formal proofs of Lemma 2.6(ii) named in Table A.1 instead write a Mackey functor as the direct limit of its Mackey subfunctors whose values are finitely generated, and show that higher limits commute with this direct limit. The construction of these subfunctors is shared. The other proof and the Rocq declaration listed for Lemma 2.6(ii) also use the isomorphisms, in each degree, between the cohomology of the cochain complex induced by the bar resolution and the higher limits.

Let M be a simple Mackey functor and let Q be a minimal subgroup for M. The other proof of Lemma 2.8(ii) uses simplicity and minimal subgroups, without first constructing the simple Mackey functor $S_{Q,M(Q)}$. The elements whose restrictions to the $\mathcal{F}$-conjugates of Q vanish form a Mackey subfunctor of M. This fact is used by the formal proofs of Lemma 2.8(ii) named in Table A.1, by the other proof and by the Rocq declaration listed for this lemma. When the central subgroup Z of Proposition 3.2 is contained in Q, the formal proofs of this proposition named in that table take the conclusion of Lemma 2.8(ii) from the fact that the Mackey subfunctor formed by these elements is zero. The Rocq declaration listed for Lemma 2.8(ii) and the other proof of this lemma also rest on the vanishing of this subfunctor. Like the formal proofs of Lemma 2.8(ii) named in that table, the other proof of Lemma 2.8(iii) uses the generation of the values of M by transfers from the $\mathcal{F}$-conjugates of Q and two identities for restriction after transfer. For $U \leqslant S$ and an $\mathcal{F}$-conjugate L of Q contained in U, restriction to L after transfer from L is the relative trace from L to $N_U(L)$ on $M(L)$. Restriction to L after transfer from an $\mathcal{F}$-conjugate of Q contained in U is zero if that conjugate is not U-conjugate to L. The other proof of Lemma 2.9 combines the reduction from arbitrary to simple Mackey functors with the vanishing of $\varprojlim^i_{\mathcal{O}(\mathcal{F}^c)} N^*$ for every $i > 0$ and every simple Mackey functor N for $\mathcal{F}$ over $\mathbb{F}_p$ whose minimal subgroups are $\mathcal{F}$-centric. The Rocq declaration listed for Lemma 2.9 uses this vanishing as well. The formal counterparts of Lemma 4.1 are not listed above, since the formal proofs of Theorem 1.1 depend on them, as does the other proof of Proposition 4.3. For Proposition 3.2, the other proof uses the vanishing of the higher limits of every functor on $\mathcal{O}(\mathcal{F}^c)$ that is zero outside the $\mathcal{F}$-conjugacy class of an $\mathcal{F}$-centric subgroup P with $O_p(\mathrm{Out}_{\mathcal{F}}(P)) \neq 1$. The other proof of Lemma 4.2 combines the vanishing of restrictions in Lemma 2.8(iii) with Alperin's fusion theorem in its form for the orbit category, and the Rocq declaration listed for Lemma 4.2 uses the theorem in its form for centric subgroups. The other proof of Proposition 4.3 uses this proof of Lemma 4.2 and the hypothesis that $\varprojlim^i_{\mathcal{O}(C_{\mathcal{F}}(E)^c)} N^* = 0$ for every $i > 0$, every E in $\mathcal{F}^e$ and every Mackey functor N for $C_{\mathcal{F}}(E)$ over $\mathbb{F}_p$. Induction on $|S|$ then gives Proposition 5.1, followed by Theorem 1.1 and Corollary 1.2. These proofs and the formal proofs of Theorem 1.1 share the construction of the Mackey functor for $\mathcal{F}/Z$ in the proof of Proposition 3.2, the vanishing of higher limits used by the other proof of this proposition, the deduction of Theorem 1.1 over $\mathbb{Z}_{(p)}$ by Nakayama's lemma and the vanishing for simple Mackey functors whose minimal subgroups are $\mathcal{F}$-centric. They also share the isomorphisms, in each degree, between the cohomology of the cochain complex

induced by the bar resolution and the higher limits. The other proofs of Lemmas 2.6(ii) and 2.9 use these isomorphisms as well.

C.3 Long exact sequences of higher limits and composition factors

For a Mackey functor M and a Mackey subfunctor $N \leqslant M$, the long exact sequence of higher limits associated with $0 \to N \to M \to M/N \to 0$ is proved in both formalisations. The connecting homomorphism is constructed, exactness at all three consecutive terms is proved in every degree, and the first map is proved injective in degree 0. The coefficient ring may be any commutative ring, and the values may be arbitrary modules. Saturation is not needed. In Rocq, for every functor Φ from a category $\mathcal{D}$ to $\mathcal{O}(\mathcal{F})$, the long exact sequence is also proved for the higher limits over $\mathcal{D}$ of $N^* \circ \Phi^{\mathrm{op}}$, $M^* \circ \Phi^{\mathrm{op}}$ and $(M/N)^* \circ \Phi^{\mathrm{op}}$.

Formalisation.

In Lean these statements are `Sharpness.FusionSystem.V29.compositionDerived_longExact` and `Sharpness.FusionSystem.V29.compositionDerivedInclusion_zero_mono`, and in Rocq they are `DPCompositionExact.subquotient_higher_limit_long_exact` and `DPCompositionExact.subquotient_degree_zero_inclusion_injective`. Instead of the Lean long exact sequence of higher limits, the Lean formal proof of Theorem 1.1 uses `higherLimit_map_isIso_of_kernel_acyclic`, the statement described in Chapter 2 on surjective maps of functors whose kernel has zero higher limits in positive degrees. Its proof rests on `Sharpness.coefficientCochain_shortExact`, by which the cochain complexes induced by the bar resolution with coefficients in the terms of a short exact sequence of functors form a short exact sequence. The proof also uses the long exact cohomology sequence of a short exact sequence of cochain complexes, from Mathlib, and the natural isomorphism between the cohomology of these cochain complexes and the higher limits. In Rocq, the proof of `subquotient_derived_extension_zero` uses the long exact sequence of higher limits directly.

Let M be a Mackey functor with a given composition series. It is also proved in both formalisations that every factor of this series is isomorphic, as a Mackey functor, to the simple Mackey functor constructed from the value of the factor at a minimal subgroup Q. The isomorphism respects restrictions and transfers, not only the values at objects. In the Lean statement, the field is $\mathbb{F}_p$ and Q is a subgroup of minimal order at which the factor is nonzero, chosen in the formalisation. In the Rocq statement, the field is arbitrary and Q is any given minimal subgroup. The Lean isomorphism `Sharpness.FusionSystem.V29.compositionFactorIdentification` is not used by the Lean formal proof of Theorem 1.1. For the Rocq statement, the saved Rocq dependency sets do not establish non-use.

Formalisation.

The Rocq declaration is `DPCompositionIdentification.factor_identification`.

Appendix D

Logical foundations and sources

The source of a mathematical statement need not be a dependency of its formal proof. A definition can follow a book, while its properties are proved from that definition in the formalisations. A theorem can follow an argument in an article without the article's conclusion becoming an assumption. At the bottom of the formal dependency chain are the rules of the logical system and the axioms listed in Chapter 1.

D.1 Sources and formal proofs

Table D.1 distinguishes the source of a mathematical statement from its proof in the formalisations. A work may be cited for a definition, a proof method or an earlier form of a result. Where no cited source is established, the table names only the construction in the formalisations or in the libraries.

Table D.1: Sources and formal proofs of the statements and constructions.

Statement or construction	Cited source or method	Formal proof and hypotheses
Fusion systems, saturation, central subgroups and orbit categories	Aschbacher, Kessar and Oliver [2, Definitions I.2.2 and II.5.3, Proposition I.2.5 and Lemma II.5.5]. The documentation of `Sharpness.FusionSystem.AKOCentralizerInput` refers to Aschbacher, Kessar and Oliver [2, Definition I.5.3 and Theorem I.5.5] for the centraliser and its saturation.	The categories and quotients are defined in the formalisations. The formal definition of central subgroups is proved equivalent to that of Definition 2.2, and the saturation of centralisers and central quotients is proved. In both formalisations, saturation is expressed by the Sylow and extension properties, whereas Aschbacher, Kessar and Oliver define it by the condition that every subgroup of S is $\mathcal{F}$-conjugate to a subgroup that is fully automised and receptive. The equivalence of the two definitions is not formalised, and no formal equivalence with every other historical definition is claimed.

Statement or construction	Cited source or method	Formal proof and hypotheses
Higher limits and Lemma 2.3	The isomorphism of the higher limits with the cohomology of the cochain complex induced by the bar resolution is described by Aschbacher, Kessar and Oliver [2, Proposition III.5.3], citing Gabriel and Zisman [17, Appendix II, Proposition 3.3]. In the manuscript, Lemma 2.3 is attributed to Jackowski and McClure [30, Lemma 3.1], and Jackowski, McClure and Oliver [31, Proposition 5.1(ii)] stated it for functors to abelian groups.	In Lean, higher limits are Ext groups from the constant functor, and the proofs compute them from projective resolutions of this functor. The Rocq proofs use injective resolutions. The natural isomorphisms of the higher limits with the cohomology of the cochain complex induced by the bar resolution, and the isomorphisms of Lemma 2.3, are constructed. In both systems, a contravariant functor on a category is written as a functor on its opposite category.
Mackey functors and Lemma 2.6	Díaz and Park [13, Definition 2.1]. The generators in Lemma 2.6(i) are obtained from an element by restriction, conjugation and transfer, and there are finitely many of them. The argument of Aschbacher, Kessar and Oliver [2, Proposition III.5.5(b)] shows that higher limits over a category with finitely many objects and morphisms commute with the direct limit of the subfunctors whose values are finitely generated.	The Mackey formula for inclusions and its form for all morphisms of $\mathcal{F}$ are proved equivalent. The values of the Mackey subfunctor generated by a finite family of elements are finitely generated, since S is finite. The terms of the cochain complex induced by the bar resolution for the orbit category are finite products, and filtered colimits commute with them. The orbit category is finite, whereas the filtered indexing category may be infinite. The coefficient ring need not be noetherian.
Simple Mackey functors and Lemma 2.8	Díaz and Park [13, Propositions 3.1–3.2 and formulas (4),(6)] adapted the arguments of Webb [52, Proposition 2.1 and Section 2]. Over $\mathbb{F}_p$, Grazian and Marmo [21, Lemma 2.9] and Díaz and Park [13, Corollary 3.4] gave analogues for fusion systems of a result of Webb [52, Proposition 2.8]. The proof of Lemma 2.8(iii) uses the lemma of Grazian and Marmo.	For a given simple Mackey functor M with a minimal subgroup Q and for each $U \leqslant S$, the decomposition of $M(U)$ into images of relative traces is constructed. It has one summand for each U-conjugacy class of $\mathcal{F}$-conjugates of Q contained in U. Restriction to a representative L of such a class is proved to be injective on the summand for L and zero on the other summands, and Lemma 2.8(ii) is established. It is also proved, for Lemma 2.8(iii), that in characteristic p the relative trace from L to $N_T(L)$ on the value at L of the given simple Mackey functor is zero for subgroups $L \leqslant T$ of S with $C_T(L) \not\leqslant L$. The proof uses the subgroup $LC_T(L)$, whose quotient by L is a nontrivial p-group that acts trivially on this value. No classification theorem for all simple Mackey functors is assumed.

Statement or construction	Cited source or method	Formal proof and hypotheses
The vanishing theorem for $\mathcal{F}^c$-restricted Mackey functors and Lemma 2.9	Díaz and Park [13, Theorem A and Proposition 4.3] used Jackowski and McClure [30, Corollary 5.16], and Díaz and Libman [12, Proposition 2.9 and Remark 2.10]. Díaz and Libman cited Puig [48, Proposition 4.7].	It is proved that the product of two representable functors on $\mathcal{O}(\mathcal{F}^c)$ is a finite coproduct of representable functors, as in the proof of Puig's result that the additive extension of $\mathcal{O}(\mathcal{F}^c)$ has products. The vanishing theorem for $\mathcal{F}^c$-restricted Mackey functors is also proved, over $\mathbb{Z}_{(p)}$ in `Sharpness.FusionSystem.restrictedCentric_acyclic` and over every commutative $\mathbb{Z}_{(p)}$-algebra in `DiazPark.FusionCentricRestrictedAcyclicity.DPFusionCentricRestrictedAcyclicity.restricted_centric_higher_limits_zero`. Lemma 2.9 concerns Mackey functors over $\mathbb{F}_p$, and its formal counterparts are proved over $\mathbb{F}_p$ in Lean and over every field of characteristic p in Rocq. In their proofs, the reduction for Mackey functors whose values are finite dimensional uses composition series and long exact sequences. The general case then follows by Lemma 2.6(ii), since every Mackey functor is the direct limit of its Mackey subfunctors whose values are finitely generated.
Central quotients, Lemma 3.1 and Proposition 3.2	Let $\mathcal{F}$ be a saturated fusion system over a finite p-group S, let $Z \leqslant S$ be central in $\mathcal{F}$, and let $\mathcal{X}$ be the set of subgroups $P \leqslant S$ that contain Z and for which P/Z is $\mathcal{F}/Z$-centric. Broto, Castellana, Grodal, Levi and Oliver [4, Lemma 6.4(a)] proved $\mathcal{X} \subseteq \mathcal{F}^c$. The proof of the converse in [4, Corollary 6.14, p. 3856] shows that $O_p(\mathrm{Out}_{\mathcal{F}}(P)) \neq 1$ for every $\mathcal{F}$-centric subgroup P outside $\mathcal{X}$, and hence that $\mathcal{F}^{cr} \subseteq \mathcal{X}$. That proof assumes that $\mathcal{F}/Z$ is the fusion system of a finite group, but the arguments from it used in the manuscript require only that $\mathcal{F}$ is saturated and Z is central in $\mathcal{F}$. The isomorphism of $\mathcal{O}^{\mathcal{X}}(\mathcal{F})$ with $\mathcal{O}((\mathcal{F}/Z)^c)$ is established in [4, p. 3856, equation (3)]. By the argument of Broto, Levi and Oliver [6, proof of Corollary 3.6], restriction from $\mathcal{O}(\mathcal{F}^c)$ to $\mathcal{O}^{\mathcal{X}}(\mathcal{F})$ preserves the higher limits of functors from $\mathcal{O}(\mathcal{F}^c)^{\mathrm{op}}$ to $\mathbb{Z}_{(p)}$-modules. Since higher limits are invariant under a change of rings, the same holds for functors to $\mathbb{F}_p$-modules.	The four properties of $\mathcal{X}$ in Lemma 3.1(i) and the isomorphism of orbit categories in Lemma 3.1(ii) are proved. Proposition 3.2 then follows as in the proof in the manuscript, by reduction to a simple Mackey functor with a minimal subgroup Q and by treating the two cases $Z \not\leqslant Q$ and $Z \leqslant Q$. Its hypothesis on the quotient holds by the induction hypothesis in the proof of Proposition 5.1. No hypothesis of the formal statements requires $\mathcal{F}$ or $\mathcal{F}/Z$ to be the fusion system of a finite group.

Statement or construction	Cited source or method	Formal proof and hypotheses
Centralisers, Lemmas 4.1 and 4.2 and Proposition 4.3	For $S \neq 1$, Yalçın [53, Lemma 9.3 and Theorem 9.4] constructed a spectral sequence from which Lemma 4.1 follows. The manuscript's algebraic proof uses pairs of subgroups as Broto, Levi and Oliver [6, proof of Theorem 2.6] did, with morphisms of the orbit category in place of those of the linking system. The proof of Lemma 4.2 uses Alperin's fusion theorem [10, Theorem 4.52].	As in the proof in the manuscript, the formal proofs of Lemma 4.1 use the category $\mathcal{P}_0$ of pairs (P, E) of an $\mathcal{F}$-centric subgroup P and a subgroup $E \leqslant Z(P)$ in $\mathcal{F}^e$, and the right Kan extension along the functor $\tau\colon \mathcal{P}_0^{\mathrm{op}} \to \mathcal{F}^e$ that sends (P, E) to E. For each E in $\mathcal{F}^e$, let $\mathcal{P}_E$ be the subcategory of $\mathcal{P}_0$ whose objects are the pairs (P, E) and whose morphisms restrict to the identity on E. The value at E of this extension of a functor Y is proved isomorphic, naturally in Y, to the inverse limit of the restriction of Y to $\mathcal{P}_E$. The formal counterparts of Lemma 4.1 assume $S \neq 1$. In the formal proofs of Proposition 4.3, they are used only for simple Mackey functors whose minimal subgroups are not $\mathcal{F}$-centric, and there are no such functors when $S = 1$. In both systems, Alperin's fusion theorem in its form for centric subgroups is proved from saturation. The Rocq proof of Lemma 4.2 uses this form, whereas the Lean proof uses a separately proved form for the orbit category.
Proposition 5.1 and Theorem 1.1	In the manuscript, Proposition 5.1 is proved by induction through central quotients and centralisers, and Theorem 1.1 follows by the finite generation of higher limits, Nakayama's lemma and Lemma 2.6(ii).	In both formalisations, the centralisers and their quotients used in the induction on $\lvert S\rvert$ are constructed as fusion systems and proved to be saturated, and each quotient is proved to lie over a p-group of smaller order than S. In Lean, Nakayama's lemma from Mathlib is used, in the form `Submodule.eq_bot_of_le_smul_of_le_jacobson_bot`. In Rocq, the case of Nakayama's lemma that is needed, for multiplication by p on a finitely generated $\mathbb{Z}_{(p)}$-module, is proved by eliminating its generators one at a time. No result from the literature is a hypothesis of the formal counterparts of Theorem 1.1 and Corollary 1.2.

Statement or construction	Cited source or method	Formal proof and hypotheses
Corollary 1.2 and the stable elements theorem	In the manuscript, Corollary 1.2 follows from Theorem 1.1, since, for every $n \geqslant 0$, $H^n(-;\mathbb{F}_p)$ is the restriction to $\mathcal{O}(\mathcal{F}^c)$ of the contravariant part of a Mackey functor for $\mathcal{F}$ over $\mathbb{F}_p$. By the stable elements theorem of Broto, Levi and Oliver [6, Theorem 5.8], for a centric linking system $\mathcal{L}$ associated to $\mathcal{F}$, $H^n(\lvert\mathcal{L}\rvert;\mathbb{F}_p)$ is the inverse limit of $H^n(-;\mathbb{F}_p)$ over $\mathcal{O}(\mathcal{F}^c)$.	The cohomology groups, with restriction and transfer, form a Mackey functor. For a given centric linking system $\mathcal{L}$, the formal proof of the stable elements theorem for $\lvert\mathcal{L}\rvert$ without p-completion uses the augmentation of the subgroup decomposition, which is proved to be a quasi-isomorphism.
Excision and the homology of geometric realisations	Constructions in the formalisations, using subdivision of singular chains and chains that are small with respect to a cover.	The relative excision map and the chain map from the simplicial chains of a simplicial set to the singular chains of its geometric realisation are proved to induce homology isomorphisms. The Rocq statements are quantified over `Real : realType`.
Sharpness of the centraliser decomposition	The functor $E \mapsto \lvert\overline{C}_{\mathcal{L}}(E)\rvert$ of the centraliser decomposition is from Broto, Levi and Oliver [6, Definition 2.4, Proposition 2.5(c) and Theorem 2.6]. The implication from the sharpness of the subgroup decompositions is an observation of Yalçın [53, §9, p. 55].	The declarations `Sharpness.FusionSystem.CentricLinkingSystem.BLO_centralizer_decomposition_sharp` and `DiazParkBLO.BLOEdgeFinal.DPBLOEdgeFinal.centralizer_cohomological_sharpness` of the supplement on the centraliser decomposition have $S \neq 1$ and a centric linking system associated to $\mathcal{F}$ among their hypotheses. The functor $E \mapsto \lvert\overline{C}_{\mathcal{L}}(E)\rvert$ and the augmentation of the centraliser decomposition are constructed. For every $n \geqslant 0$, the functor $E \mapsto H^n(\lvert\overline{C}_{\mathcal{L}}(E)\rvert;\mathbb{F}_p)$ is proved to be naturally isomorphic to the right Kan extension along τ of the functor that sends a pair (P, E) to $H^n(P;\mathbb{F}_p)$. In the supplement, the Rocq proof depends on constructed spectral sequences of filtered cochain complexes and on their proved convergence and collapse. It reuses the isomorphism of the stable elements theorem and the formula for its components from the Rocq formalisation, and this isomorphism is obtained from the collapse of such a spectral sequence for the subgroup decomposition. The conclusion concerns cohomology, not a homotopy equivalence.

Statement or construction	Cited source or method	Formal proof and hypotheses
Counterexample over the rational numbers	The counterexample in item (i) of Section 3.2.1 concerns the coefficient hypotheses of Díaz and Park [13, Corollary 3.4]. The formal proofs of Lemma 2.8(iii) instead use the vanishing in characteristic p of the relative trace from L to $N_T(L)$ on $M(L)$, for a simple Mackey functor M and subgroups $L \leqslant T$ of S with $C_T(L) \not\leqslant L$.	It is proved that $H^0(-;\mathbb{Q})$, with its restriction, transfer and conjugation maps, is simple, has unique minimal subgroup 1 and is nonzero at S, although $C_S(1) \not\leqslant 1$ when $S \neq 1$. A saturated fusion system over the group of order 2 is also constructed as an example. The counterexample does not contradict Theorem 1.1.

The vanishing theorem for $\mathcal{F}^c$-restricted Mackey functors rests, in the works cited for it, on the theorem of Jackowski and McClure on proto-Mackey functors and, through Díaz and Libman, on Puig's result that the additive extension of $\mathcal{O}(\mathcal{F}^c)$ has products. For simple Mackey functors, the cited arguments go back to Webb, through Díaz and Park and through Grazian and Marmo. All of these arguments are reconstructed in the formal proofs, which assume no cited result.

D.2 Formal dependencies

Figures D.1 and D.2 group the formal dependencies by mathematical purpose. An arrow denotes a dependence through one or more proved steps. It does not assert that every declaration of the node above the arrow directly uses every declaration of the node below it. The saved Lean dependency graph distinguishes types, bodies and the links between inductive types, their constructors and the computation rules of their recursors. The Rocq evidence consists of the declarations named in proof scripts, the saved Rocq dependency sets and the module imports. These three kinds of Rocq evidence are not interchangeable.

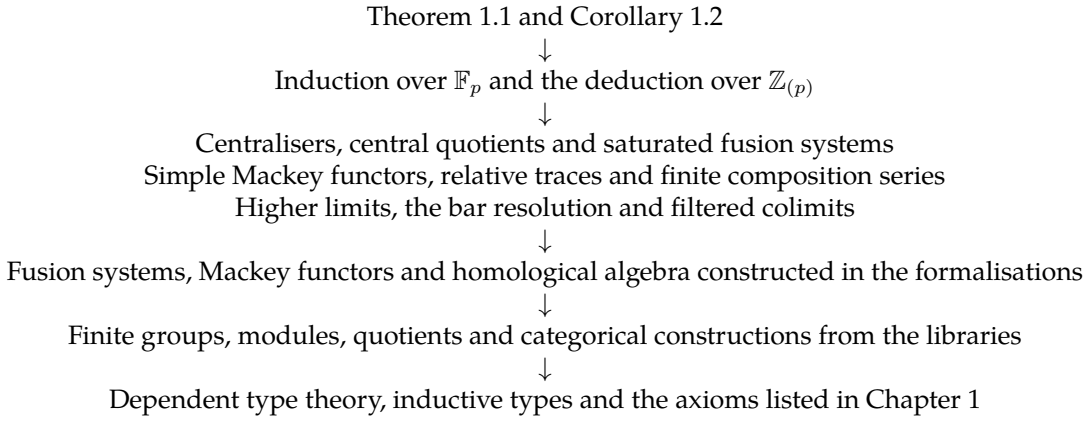


Figure D.1: Dependencies of the formal proofs of Theorem 1.1 and Corollary 1.2, grouped by mathematical purpose. The saved Lean dependency graph also contains library dependencies, including topology and analysis.

The further dependencies of the supplement, which starts from a given centric linking system and its geometric realisation, are shown in Figure D.2. The main formal theorem of the supplement is proved in every degree under the hypothesis $S \neq 1$. The formal proofs of the counterexample over the rational numbers in item (i) of Section 3.2.1 rest on the proved

restriction and transfer formulas for $H^0(-;\mathbb{Q})$. The simplicity of this Mackey functor and its minimal subgroup are obtained from these formulas.

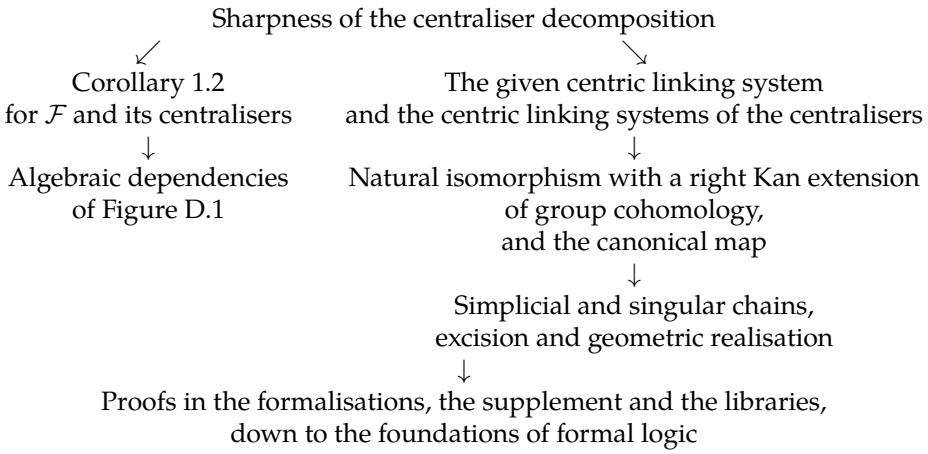


Figure D.2: Dependencies of the formal proofs of the sharpness of the centraliser decomposition in the supplement. Their conclusion concerns cohomology and the augmentation of the centraliser decomposition, not a homotopy equivalence or a result after p-completion.

D.3 Constructions in the libraries and in the formalisations

In Lean, `Sharpness.higherLimit` is expressed as Ext from the constant functor whose value is the coefficient ring. Mathlib's `Ext`, in `Mathlib.CategoryTheory.Abelian.Ext`, is constructed as the derived functors of the linear Hom bifunctor in its contravariant variable. The existence of enough projectives in the category of functors is proved in `Sharpness.Unconditional.moduleDiagramEnoughProjectives`, and the projective bar resolution `Sharpness.Unconditional.categoryBarResolution` of this constant functor is constructed in the Lean formalisation. Computing Ext from this resolution gives the natural isomorphism `Sharpness.Unconditional.categoryBarExtNatIso` between the higher limits and the cohomology of the cochain complex induced by the bar resolution.

In Rocq, `DiazPark.CategoryDerivedLimits.derived_limit` is the cohomology of the inverse limit of a chosen injective resolution. Enough injectives are constructed by embedding the underlying abelian group in a divisible abelian group and coinducing from $\mathbb{Z}$ to the coefficient ring. The isomorphism `DiazPark.BarDerivedComparison.DiazParkBarDerivedComparison.bar_derived_limit_iso` between the cohomology of the cochain complex induced by the bar resolution and the higher limits is constructed. The infrastructure for finite groups and modules comes from Mathematical Components. The categories of functors, the resolutions and the chain maps between resolutions are constructed in the Rocq formalisation, and the proofs of sharpness are written there.

Higher limits are proved to commute with filtered colimits in both systems, in different generality. In Lean the indexing categories are small filtered categories, and in Rocq they are directed posets, which suffice in the proof of Lemma 2.6(ii) for the family of Mackey subfunctors whose values are finitely generated. This commutation is proved in Rocq first for the cohomology of the cochain complex induced by the bar resolution, by a direct calculation. The isomorphism of this cohomology with the higher limits defined by injective resolutions is proved by dimension shifting. The proofs of the topological consequences in Rocq use constructed spectral sequences and proved results on their pages and their convergence.

The Rocq statements that involve geometric realisation are quantified over `Real : realType`, a real closed field with the Archimedean property and the least upper bound property. In contrast, the Rocq counterpart of Theorem 1.1, `DiazParkV29.MainManuscript.DPV29DiazParkManuscript.prime_local_main`, is not quantified over `Real`. The Rocq formalisation defines `DiazParkV16b.TopologicalSimplexReals.DPStandardSimplex.real_line` to be `[the realType of (R : Type)]`, which is built from the real numbers of the standard library through the module `Rstruct` of MathComp Analysis. The module that defines it uses this instance for its standard simplices. The given linking system is an object satisfying the axioms of a centric linking system, and these axioms contain no condition on cohomology.

The module defining `real_line` is absent from the transitive project imports of `DiazParkV29.MainManuscript` and `DiazParkV29.MainCohomology`, which contain the Rocq counterparts of Theorem 1.1 and Corollary 1.2, `DiazParkV29.MainManuscript.DPV29DiazParkManuscript.prime_local_main` and `DiazParkV29.MainCohomology.DPV29GroupCohomologySharpness.prime_field_higher_limits_zero`. This conclusion follows by resolving the `Require` commands throughout the project sources. It concerns only dependence on the declarations of that module. The libraries were not expanded in this computation, so it does not exclude shared dependencies on the standard library. The saved assumption queries for these two counterparts give the three Rocq axioms listed in Chapter 1. They are not queries of `real_line`.

D.4 The logics of Lean and Rocq

In dependent type theory, a proposition is expressed by a type and a proof by a term of that type. Dependent function types express universal quantification. Inductive types, their constructors and their elimination and computation rules support the constructions of numbers, equality and recursively defined objects. Universe levels control the sizes of types and categories.

The logic of Lean is a dependent type theory with inductive types and quotient types. The union of the axioms on which the queried Lean declarations depend consists of three of the standard axioms of Lean. Propositional extensionality equates logically equivalent propositions. Classical choice selects an element from a nonempty type. Quotient soundness equates the images of related elements in a quotient. Function extensionality is derived in Lean and does not add a fourth axiom to this union. Quotients occur in groups, in the morphisms of orbit categories and in modules, and choice is also used for representatives and resolutions.

The logic of Rocq is the Calculus of Inductive Constructions. In this development, the three axioms listed in Chapter 1 are declared in the classical library of MathComp Analysis. They allow extraction of a witness from a proved existential proposition, equality of pointwise equal dependent functions, and equality of logically equivalent propositions. The quotient types in the Rocq formalisation are those of Mathematical Components. No separate axiom of quotient soundness occurs in the union of the axioms on which the queried Rocq declarations depend.

Therefore the formal chain ends at explicit logical rules and axioms, not at an unproved sharpness theorem from the literature. Checking these deductions is not a proof of consistency of the underlying logic, nor a formal verification of the executable kernel. Appendix E describes the checks with Nanoda, an independently implemented proof checker, and with comparator, and the rechecking of the compiled Rocq libraries. It also says which of these checks share definitions, an exporter or a kernel with each other or with the compilers.

D.5 Descriptions of the proof systems and the classical library

The following sources describe the proof systems and the classical library. They are separate from the bibliography, which consists of entries 1–53 of the manuscript bibliography with the same numbers. The claims of this appendix about the formalisations and the supplement come from the formal source files used for the proofs and the saved dependency and assumption queries, not from these general descriptions.

F1. de Moura and Ullrich, *The Lean 4 Theorem Prover and Programming Language*, 2021. This is a primary account of Lean's architecture. Ullrich's *An Extensible Theorem Proving Frontend*, Sections 1.3.2 and 3.2, gives further details of the type theory and kernel.

F2. The Lean development team's *Lean Language Reference*, Axioms, Propositions, and Elaboration and Compilation explain the logical principles and the distinction between elaboration and kernel checking. These online pages describe the current system. The version used for the proofs is stated in Chapter 1.

F3. The Rocq development team's *Rocq Prover Reference Manual*, Typing rules and Core language describe the calculus and checking of core terms. Its historical account relates the Calculus of Constructions of Coquand and Huet to its extension by inductive types.

F4. The MathComp Analysis source `classical/boolp.v` contains the three axioms on which the queried Rocq declarations depend. Its header credits the corresponding formulations in the Rocq standard library. It also credits Section 5 of Affeldt, Cohen and Rouhling, *Formalization Techniques for Asymptotic Reasoning in Classical Analysis*, Journal of Formalized Reasoning 11 (2018), 43–76. The article explains the design of the classical library.

Appendix E

Compilation, proof checking and size of the project sources

E.1 Compilation of the project sources

A source module is compiled after the modules on which it depends. Its compiled file is then read by later compiler invocations. Consequently, a development with many source files does not require all their proof scripts to be elaborated in a single process. The compiler still needs memory for the imported environment and the current proof. The size of that environment, the proof being elaborated and the number of concurrent processes matter more than the number of source files alone.

The formalisations contain 934 Lean modules and 1,215 Rocq modules. The supplement on the centraliser decomposition adds 23 Lean modules and 33 Rocq modules. These are project sources, separate from the libraries. In both systems, the compiled libraries were reused. The measurements of Table E.1 do not describe rebuilding Mathlib, Mathematical Components or the proof assistants from their sources. An imported library theorem has a compiled proof term. Reusing this term is different from declaring the theorem as an axiom. The kernel checks of Table E.2 also cover the library declarations in their selected dependency closures.

A separate compilation of all 2,205 project source modules, namely 957 Lean modules and 1,248 Rocq modules, was made for the measurements of this section. Each project module was compiled from source in dependency order into a separate output directory. The compiled files of the formalisations and the supplement, produced as described in Chapter 1, were not reused. The source files agreed byte for byte with those in the packages of the formalisations and the supplement. The measured times also include the diagnostic commands within the project sources, so they measure compilation of those complete files.

The computer had a processor with 16 cores and 22 logical processors. Windows 11 Pro, build 26200, reported 68,192.1 MB of physical memory. The Lean compilation ran under Windows, and the Rocq compilation ran under WSL2 with Linux 6.18.54.1 and 33,392.3 MB of guest memory. These are the conditions of this compilation.

The Lean compiler was run with `--threads=1`, an explicit `--root` directory and `-o` for each compiled file. The Rocq compiler was run with `-q`, `-async-proofs off` and `-async-proofs-j 1`, with five explicit `-Q` mappings of the project directories to logical paths. At first one Lean compiler process ran at a time. The limit rose to two concurrent processes once the check of the compiled Rocq files of the formalisations and the supplement had finished. Rocq compilation remained sequential. The Lean and Rocq compilations ran concurrently.

Quantity	Lean	Rocq
Project modules	957	1,248
Compilation elapsed time (minutes)	161.55	201.48
Sum of compiler command times (minutes)	248.26	201.45
Compiler CPU time (seconds)	14,449.63	11,773.72
Peak sampled sum of working sets or resident set sizes (MB)	7,126.1	10,485.1
Largest peak working set or resident set size of one process, reported by the operating system (MB)	3,684.6	10,484.7
Peak sampled sum of Windows private committed memory (MB)	17,063.9	—

Table E.1: Compilation of the project sources, with the compiled libraries reused.

The complete measured interval was 201.49 minutes. The Lean and Rocq compilations overlapped, so this interval is not the sum of their elapsed times. The sum of compiler command times also differs from elapsed time when commands overlap. Elapsed times include scheduling, hash comparisons, logging and observation of process completion. Other work on the computer, including the separate checker during part of the new compilation, could affect those times.

CPU time is user plus system time. Lean CPU totals concern the direct compiler processes. Rocq CPU totals are the CPU times reported by the operating system for each compiler command, together with those of the descendant processes counted for that command. Neither CPU total is a measurement of the whole computer's activity.

The Lean memory sums concern direct compiler processes only. Their working set is resident memory, whereas private committed memory is a different quantity and can be larger. The Rocq memory sum uses the sampled resident set size (RSS) of the process tree. The largest peak of one process, as reported by the operating system, is separate from either sampled sum. All MB values are decimal, with $1\,\mathrm{MB} = 10^6$ bytes.

Memory was sampled every 0.5 seconds. A brief peak can fall between samples, and sums of resident memory can count shared pages more than once. These measurements are not the peak memory use of the whole computer. Since reused process numbers could associate unrelated older processes with a compiler, the Lean memory sums use only samples taken through the direct compiler process handles. Therefore Lean memory including descendant processes is not established.

E.2 What the checks establish

Compilation and a later examination of proof terms serve different purposes. During compilation, notation, implicit arguments and proof scripts are elaborated into terms. A kernel checks the terms against their types. A later checker can examine those terms without repeating the proof scripts. Agreement between checkers strengthens the evidence for the terms checked. It does not establish that a theorem's type is the intended mathematical statement.

Table E.2 distinguishes the checks performed.

Table E.2: Proof checking and comparison.

Operation	Examination performed	Independence and limitations
Lean compilation	Elaboration of project sources and checking of resulting declarations.	Uses Lean's elaborator and kernel, with the compiled libraries.
Lean kernel replay	Selected declarations and their logical dependencies are added to an initially empty environment at trust level 0.	A fresh examination using Lean's kernel, not a second implementation of that kernel.
Nanoda	Exported Lean declarations and their dependencies are checked by a Rust implementation.	Independent kernel implementation. It uses the same definitions and exporter as the check with comparator, whereas the Lean kernel replay reads compiled declarations without that exporter.
Lean comparator	A copy of the formal statement of the Lean counterpart of Theorem 1.1, in a separate challenge file that imports modules of the Lean formalisation, is compared with the theorem in that formalisation, followed by an axiom check and a check of the exported proof by lean4checker.	One theorem only. Corollary 1.2 was not compared. The definitions are shared and are not checked independently. Its lean4checker component invokes Lean's kernel.
Rocq compilation	Project proof scripts are elaborated and the resulting declarations checked.	Uses Rocq and the compiled libraries.
`coqchk`	Compiled Rocq files and their recursive dependencies are checked without trusting the compiled proof bodies.	A separate executable that shares the Rocq kernel. It does not repeat the proof scripts.
Comparison of the two formalisations with the manuscript	Definitions, hypotheses, conclusions and named maps are compared with the manuscript.	Mathematical comparison of separately written developments, not automatic translation or a theorem equating their meanings.

In the Lean kernel replay, a theorem with a correct proof term was accepted, and the same statement with a term of the wrong type was rejected. Nanoda accepted a valid theorem, rejected an expression with the wrong type and rejected an axiom other than the Lean axioms listed in Chapter 1. For a theorem about natural numbers, a matching statement was accepted by comparator, and a changed statement and such an axiom were rejected. These controls exercise acceptance and rejection. They are not proofs of correctness or consistency of the checker programs.

The Nanoda checks used the compiled files of the formalisations and the supplement, not those of the compilation of Table E.1. The checks of the formalisations used two exports. One contained 525 selected project declarations and their dependencies, for a total of 67,161

declarations. These selected declarations include the Lean counterparts of Theorem 1.1 and Corollary 1.2. The other contained 320 further declarations selected by the Lean kernel replay, with their dependencies, for a total of 53,329. The union of the two exports, formed by declaration names, has 67,486 declarations and contains all 66,527 declarations selected by that replay. The supplement's export contained 65,385 declarations, including all 64,442 selected by its Lean kernel replay. Additional recursors account for the larger export totals. Since these closures overlap, the sum of their totals counts some declarations more than once. The closures do not denote all declarations of every project module. Exactly the Lean axioms listed in Chapter 1 were allowed. The Nanoda control rejecting an axiom outside the allowed list took place after the run on the 525 selected declarations and before the other two runs.

The runs on the supplement and on the 320 further declarations have matching hashes of the executable, the export and the configuration before and after execution. For the run on the 525 selected declarations, no such hashes of the executable were taken. That the installed executable was used in this run rests on the hash given in its installation log and on a second computation of that hash, made before the other two runs, with the same result.

The check with comparator concerned `Sharpness.FusionSystem.V29.primeLocal_sharpness`, the Lean counterpart of Theorem 1.1. The challenge file is a copy of the statement of this theorem in the Lean formalisation. It imports four modules of that formalisation, and through them the definitions that the statement uses, but not the module containing the theorem. A placeholder in the challenge file served only to state the expected proposition and was not accepted as a proof. The module containing the theorem was the solution. It passed the comparison of the statement and the declarations, the restriction of its axioms to `propext`, `Quot.sound` and `Classical.choice`, and a check by lean4checker, which uses Lean's kernel. Since the challenge file and the solution share the compiled definitions, the check with comparator shows that, for these definitions, the type of the theorem is the statement in the challenge file. It does not check the definitions themselves.

The version of comparator used was the one for Lean 4.31.0, which the formalisations also use. The checking code of comparator was called directly from a separate program, on exports made from the compiled challenge file and from the compiled files of the Lean formalisation. The comparator program itself, which builds and exports inside a sandbox, was not run, so no sandbox for untrusted programs was used. Nanoda was switched off in this check and was run separately.

The `coqchk` run passed for all 1,248 project modules. These comprise 1,215 modules of the formalisations and 33 of the supplement. The recursive closure of these modules contained 1,695 compiled Rocq files, including 447 files of the libraries. One invocation named every project module, using the installed Rocq library directory and five mappings of project directories to logical paths.

This check examined the compiled files of the formalisations and the supplement. The compilation of Table E.1 wrote its files to other directories. The checker executable and the loaded files had matching hashes before and after the run, and the project sources also had the same hashes as the corresponding sources in the packages of the formalisations and the supplement.

The check included the recursively imported compiled files, without options that skip or admit their contents. A compiled Rocq file of the libraries was accepted in a control run, and a copy with a damaged file header was rejected. This rejection shows only that a damaged file is rejected, not that a well-formed file containing an invalid proof term is rejected. A separate example in which a theorem is proved from a declared axiom was accepted, as expected. Since the `coqchk` check does not by itself restrict axioms to an allowed list, it is combined with the assumption queries and the inspection of source commands that affect checking. The

invocation did not use the option `-o` to report assumptions. Therefore the list of Rocq axioms in Chapter 1 rests on the saved assumption queries.

In this check, the checking flags stored in compiled files are respected. A separate lexical scan of the 1,248 project sources searched for a list of commands that disable checking, unchecked casts and unproved declarations, and found none of them. This scan supplements the assumption queries, but it does not determine every stored flag or inspect every command in the libraries.

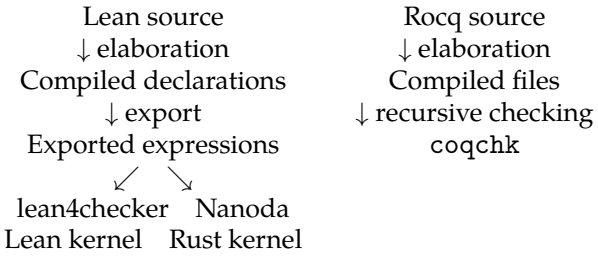


Figure E.1: Checks of proof terms. The arrows show how proof terms are processed, not implications between mathematical theorems. The Lean kernel replay of compiled declarations is a further check, not shown. For the Lean counterpart of Theorem 1.1 only, the check with comparator added a comparison of statements before lean4checker.

E.3 Time and memory of the checker runs

The measurements are given in Table E.3. Export preparation, tool installation and compilation of the checker programs are excluded. CPU seconds are user plus system CPU time. The CPU time of the check with comparator is unavailable. Elapsed seconds are the wall-clock times of the enclosing processes.

Operation	Elapsed (s)	CPU (s)	Peak MB	Measurement
Nanoda, 525 selected declarations	332.454	324.08	2,172.2	Linux maximum RSS
Nanoda, 320 further declarations	218.578	212.39	2,081.1	Linux maximum RSS
Nanoda, supplement	197.985	187.36	2,162.6	Linux maximum RSS
comparator, Theorem 1.1 only	141.699	Unavailable	3,078.7	Windows working set
`coqchk`, all project modules	5,100.358	4,951.51	7,870.4	Linux maximum RSS

Table E.3: Elapsed time, CPU time and peak memory of the checker runs.

The Nanoda and `coqchk` runs used one checking process each. The check with comparator used one Lean process with one Lean worker thread. The `coqchk` row concerns the compiled files of the formalisations and the supplement, with the recursively imported compiled files, not those of the compilation of Table E.1. The CPU and peak memory values measured with GNU time and the elapsed times of the enclosing processes cover slightly different intervals. Compilation and checking overlapped for part of the time, and the elapsed times include the effect of that load.

The export of the 525 selected declarations and their dependencies used approximately 3,851 MB of memory, measured as the Windows working set. The export for the 320 further declara-

tions has no valid peak memory measurement because its memory sampler failed. Its intended memory limit was consequently not enforced by that sampler. The checker measurement in the table was obtained separately.

The peak memory values in Table E.3 are those of single checking processes, not of the whole computer. They are given in decimal MB, with $1\,\mathrm{MB} = 10^6$ bytes.

E.4 Size of the project sources

The source counts are given in Table E.4. The line totals include blank lines and comment lines. The two language totals are kept separate because their syntax and library abstractions differ.

Project sources	Modules	Lines	Blank lines	Lines of code
Lean formalisations	934	73,010	8,385	60,172
Lean supplement	23	2,060	269	1,677
Lean combined	957	75,070	8,654	61,849
Rocq formalisations	1,215	111,752	12,840	96,924
Rocq supplement	33	3,360	329	2,933
Rocq combined	1,248	115,112	13,169	99,857

Table E.4: Size of the project sources of the formalisations and the supplement, each file counted once, without the libraries.

There are 4,567 comment lines in the combined Lean project sources and 2,086 in the combined Rocq project sources. The lines are partitioned into blank lines, comment lines and lines of code. Across both languages the combined development has 161,706 lines of code.

Adding the lines of every file in the packages would count some project sources more than once, since the package of the supplement contains that of the formalisations. The counts in Table E.4 include each selected project source once and are given separately for the formalisations and the supplement. The libraries, compiled files, logs, documentation, checker programs and control examples are excluded.

A line of code is a line containing something other than whitespace or a comment. This includes imports, definitions, theorem statements, proof scripts and diagnostic commands within these project sources. Nested comments and strings are treated according to the syntax of the language. A line containing both code and a comment is counted once, as a line of code. These counts are reproducible measures of source text, not numbers of mathematical steps.

The totals include the project sources of the results of Appendix C. They are not a minimal source count for Theorem 1.1.

Bibliography

[1] M. Aschbacher and A. Chermak, *A group-theoretic approach to a family of 2-local finite groups constructed by Levi and Oliver*, Ann. of Math. (2) **171** (2010), 881–978.

[2] M. Aschbacher, R. Kessar, and B. Oliver, *Fusion Systems in Algebra and Topology*, London Math. Soc. Lecture Note Ser., vol. 391, Cambridge Univ. Press, Cambridge, 2011.

[3] M. Aschbacher and B. Oliver, *Fusion systems*, Bull. Amer. Math. Soc. (N.S.) **53** (2016), 555–615.

[4] C. Broto, N. Castellana, J. Grodal, R. Levi, and B. Oliver, *Extensions of p-local finite groups*, Trans. Amer. Math. Soc. **359** (2007), 3791–3858.

[5] C. Broto, R. Levi, and B. Oliver, *Homotopy equivalences of p-completed classifying spaces of finite groups*, Invent. Math. **151** (2003), 611–664.

[6] C. Broto, R. Levi, and B. Oliver, *The homotopy theory of fusion systems*, J. Amer. Math. Soc. **16** (2003), 779–856.

[7] I. Chatterji (ed.), *Guido's Book of Conjectures*, Monogr. Enseign. Math., vol. 40, L'Enseignement Mathématique, Geneva, 2008.

[8] A. Chermak, *Fusion systems and localities*, Acta Math. **211** (2013), 47–139.

[9] M. Clelland and C. Parker, *Two families of exotic fusion systems*, J. Algebra **323** (2010), 287–304.

[10] D. A. Craven, *The Theory of Fusion Systems: An Algebraic Approach*, Cambridge Stud. Adv. Math., vol. 131, Cambridge Univ. Press, Cambridge, 2011.

[11] D. A. Craven, B. Oliver, and J. Semeraro, *Reduced fusion systems over p-groups with abelian subgroup of index p: II*, Adv. Math. **322** (2017), 201–268.

[12] A. Díaz and A. Libman, *The Burnside ring of fusion systems*, Adv. Math. **222** (2009), 1943–1963.

[13] A. Díaz and S. Park, *Mackey functors and sharpness for fusion systems*, Homology Homotopy Appl. **17** (2015), 147–164.

[14] A. Díaz, A. Ruiz, and A. Viruel, *All p-local finite groups of rank two for odd prime p*, Trans. Amer. Math. Soc. **359** (2007), 1725–1764.

[15] W. G. Dwyer, *Sharp homology decompositions for classifying spaces of finite groups*, in *Group Representations: Cohomology, Group Actions and Topology*, Proc. Sympos. Pure Math., vol. 63, Amer. Math. Soc., Providence, RI, 1998, 197–220.

[16] W. G. Dwyer and C. W. Wilkerson, *A cohomology decomposition theorem*, Topology **31** (1992), 433–443.

[17] P. Gabriel and M. Zisman, *Calculus of Fractions and Homotopy Theory*, Ergeb. Math. Grenzgeb., vol. 35, Springer, Berlin, 1967.

[18] G. Glauberman and J. Lynd, *Control of fixed points and existence and uniqueness of centric linking systems*, Invent. Math. **206** (2016), 441–484.

[19] G. Glauberman and J. Lynd, *Cohomology on the centric orbit category of a fusion system*, Fund. Math. **269** (2025), 187–199.

[20] D. Gorenstein, *Finite Groups*, 2nd ed., Chelsea, New York, 1980.

[21] V. Grazian and E. Marmo, *Sharpness of saturated fusion systems on a Sylow p-subgroup of* $\mathrm{G}_2(p)$, Homology Homotopy Appl. **25** (2023), 329–342.

[22] V. Grazian and C. Parker, *Saturated fusion systems on p-groups of maximal class*, Mem. Amer. Math. Soc. **307** (2025), no. 1549.

[23] V. Grazian, C. Parker, J. Semeraro, and M. van Beek, *Fusion systems related to polynomial representations of* $\mathrm{SL}_2(q)$, J. London Math. Soc. (2) **113** (2026), Art. e70481.

[24] J. Grodal, *Higher limits via subgroup complexes*, Ann. of Math. (2) **155** (2002), 405–457.

[25] J. Grodal and S. D. Smith, *Propagating sharp group homology decompositions*, Adv. Math. **200** (2006), 525–538.

[26] E. Henke, *Subcentric linking systems*, Trans. Amer. Math. Soc. **371** (2019), 3325–3373.

[27] E. Henke, A. Libman, and J. Lynd, *Punctured groups for exotic fusion systems*, Trans. London Math. Soc. **10** (2023), 21–99.

[28] E. Henke and J. Lynd, *Fusion systems with Benson–Solomon components*, Duke Math. J. **171** (2022), 673–737.

[29] E. Henke and S. Shpectorov, *Fusion systems over Sylow p-subgroups of* $\mathrm{PSp}_4(p^a)$, unpublished manuscript.

[30] S. Jackowski and J. McClure, *Homotopy decomposition of classifying spaces via elementary abelian subgroups*, Topology **31** (1992), 113–132.

[31] S. Jackowski, J. McClure, and B. Oliver, *Homotopy classification of self-maps of BG via G-actions. Part II*, Ann. of Math. (2) **135** (1992), 227–270.

[32] J. Lannes, *Sur les espaces fonctionnels dont la source est le classifiant d'un p-groupe abélien élémentaire*, Inst. Hautes Études Sci. Publ. Math. **75** (1992), 135–244.

[33] R. Levi and B. Oliver, *Construction of 2-local finite groups of a type studied by Solomon and Benson*, Geom. Topol. **6** (2002), 917–990; and *Correction to: Construction of 2-local finite groups of a type studied by Solomon and Benson*, Geom. Topol. **9** (2005), 2395–2415.

[34] A. Libman, *The normaliser decomposition for p-local finite groups*, Algebr. Geom. Topol. **6** (2006), 1267–1288.

[35] S. Mac Lane, *Categories for the Working Mathematician*, 2nd ed., Grad. Texts in Math., vol. 5, Springer, New York, 1998.

[36] B. Oliver, *Existence and uniqueness of linking systems: Chermak's proof via obstruction theory*, Acta Math. **211** (2013), 141–175.

[37] B. Oliver, *Simple fusion systems over p-groups with abelian subgroup of index p: I*, J. Algebra **398** (2014), 527–541.

[38] B. Oliver and A. Ruiz, *Reduced fusion systems over p-groups with abelian subgroup of index p: III*, Proc. Roy. Soc. Edinburgh Sect. A **150** (2020), 1187–1239.

[39] B. Oliver and A. Ruiz, *Simplicity of fusion systems of finite simple groups*, Trans. Amer. Math. Soc. **374** (2021), 7743–7777.

[40] C. Parker and J. Semeraro, *Fusion systems over a Sylow p-subgroup of* $\mathrm{G}_2(p)$, Math. Z. **289** (2018), 629–662.

[41] C. Parker and J. Semeraro, *Fusion systems on maximal class 3-groups of rank two revisited*, Proc. Amer. Math. Soc. **147** (2019), 3773–3786.

[42] C. Parker and G. Stroth, *A family of fusion systems related to the groups* $\mathrm{Sp}_4(p^a)$ *and* $\mathrm{G}_2(p^a)$, Arch. Math. (Basel) **104** (2015), 311–323.

[43] M. Praderio Bova, *Green correspondence on centric Mackey functors over fusion systems*, J. Algebra **638** (2024), 279–357.

[44] M. Praderio Bova, *Computing higher limits over the fusion orbit category via amalgams*, arXiv:2601.14983v1, 21 January 2026.

[45] M. Praderio Bova, *An inductive approach to the Díaz–Park sharpness conjecture*, arXiv:2604.21161v1, 23 April 2026.

[46] L. Puig, *Structure locale dans les groupes finis*, Mém. Soc. Math. France **47** (1976), 5–132.

[47] L. Puig, *La classification des groupes finis simples: bref aperçu et quelques conséquences internes*, Séminaire Bourbaki, exp. no. 584, Astérisque **92–93** (1982), 101–128.

[48] L. Puig, *Frobenius categories*, J. Algebra **303** (2006), 309–357.

[49] L. Puig, *Frobenius Categories versus Brauer Blocks: The Grothendieck Group of the Frobenius Category of a Brauer Block*, Progr. Math., vol. 274, Birkhäuser, Basel, 2009.

[50] A. Ruiz and A. Viruel, *The classification of p-local finite groups over the extraspecial group of order p^3 and exponent p*, Math. Z. **248** (2004), 45–65.

[51] M. van Beek, *Exotic fusion systems related to sporadic simple groups*, Math. Proc. Cambridge Philos. Soc. **178** (2025), 331–384.

[52] P. Webb, *Two classifications of simple Mackey functors with applications to group cohomology and the decomposition of classifying spaces*, J. Pure Appl. Algebra **88** (1993), 265–304.

[53] E. Yalçın, *Higher limits over the fusion orbit category*, Adv. Math. **406** (2022), Art. 108482.

Further references

[R1] K. S. Brown, *Cohomology of Groups*, Grad. Texts in Math., vol. 87, Springer, New York, 1982.